



AISSMS
INSTITUTE OF INFORMATION TECHNOLOGY
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A
PROJECT REPORT
ON

**Design and Development of Autonomous Dual-Drone System for Survivor
Detection and Aid Delivery in Flood Zones**

SDG 11: Sustainable Cities and Communities

SUBMITTED TO AISSMS'S INSTITUTE OF INFORMATION TECHNOLOGY,
PUNE IN THE PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE OF

BACHELOR OF TECHNOLOGY IN
ELECTRONICS AND TELECOMMUNICATION
BY

Nirmity Puslekar	22310100
Sumit Rathod	22310101
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UNDER THE GUIDANCE OF

Prof. Rahul N. Jadhav

**DEPARTMENT OF ELECTRONICS AND
TELECOMMUNICATION**

BY

**ALL INDIA SHRI SHIVAJI MEMORIAL SOCIETY'S
INSTITUTE OF INFORMATION TECHNOLOGY, PUNE
411001**

ACADEMIC YEAR: 2025-26



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CERTIFICATE

This is to certify that Project Report entitled

Design and Development of Autonomous Dual-Drone System for Survivor Detection and Aid Delivery in Flood Zones

Submitted
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This is the record of bonafide work carried out by them in partial fulfillment of the requirement for the award of the Degree of **Bachelor Technology (Electronics and Telecommunication)**, as prescribed by the AISSMS IOIT in the Academic Year 2025-2026.

This project report has not been earlier submitted to any other Institute or University for the award of any degree.

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ABSTRACT

Floods are one of the maximum devastating herbal disasters, often inflicting huge destruction, displacement and loss of existence. Rapid detection of survivors and timely transport of important help is critical to minimize injuries. This project offers the design and development of an autonomous twin drone system capable of locating survivors and presenting help in flood-affected regions. The system integrates two specialized drones: Drone A for aerial surveillance and Drone B for payload transport. Drone A is prepared with a thermal imaging digital camera and GPS module that autonomously scans flooded regions, detects human warmness signatures and marks particular survivor coordinates using photo processing and AI-based detection algorithms. These coordinates are then transmitted wirelessly to Drone B, which autonomously navigates to the region and promises emergency materials which include food, water and medical kits. Both drones communicate via a stable IoT-based control network for real-time tracking and coordination. This device focuses on autonomy, accuracy, and efficiency while reducing human intervention during rescue operations. Simulation and prototype studies prove that the proposed structure of a twin-drone could enhance response times, operational reliability, and overall performance in disaster control cases.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Project

Floods are one of the most frequent and devastating natural disasters, causing extreme loss of life, property, and infrastructure. In such situations, harsh environmental conditions, submerged terrain, and communication breakdowns make it a major challenge for rescue teams to quickly reach stranded survivors and offer essential help. Unmanned aerial vehicles (UAVs) offer the hope of conducting search operations instead of sending personnel to flooded areas. Recent advances in UAVs, artificial intelligence, and wireless communication technology have opened new opportunities for effective disaster response by providing aerial surveillance and real-time data collection in regions inaccessible to ground vehicles.

1.2 Problem Statement

Despite the availability of several solutions, existing single UAV systems are unable to achieve the dual functionality of carrying heavy payloads for high-number aerial surveillance and delivery due to rapid battery drain and high latency. When a single drone must perform computationally intensive AI-based detection alongside delivery tasks for heavy lift payloads, battery life and flight endurance are significantly compromised. Furthermore, manual search operations are limited by the fatigue factor, which causes latency and means that survivors can remain undetected for significant periods of time. Currently, no framework in the literature provides a coordinated multi-drone framework that can perform edge-computed detection and autonomous aid delivery without the need for a ground station.

1.3 Motivation / Need for the Project

During flood crises, the golden time to seek medical help is frequently wasted due to difficult terrain and the inherent dangers of ground-based rescue. Therefore, there is a growing need for self-sustaining aerial structures capable of acting in search, detection, and aid transport operations with minimal human intervention. The motivation for this project is to propose a decentralized architecture based on a "scout-courier" framework to overcome these operational limitations. This dual-drone approach increases operational efficiency by providing simultaneous detection and transport, reducing reaction time, and increasing the chances of saving lives during flooding

1.4 Objectives of the Project

The main objectives of this project are:

- To create two autonomous drones, one for delivering aid in flood-prone areas and the other for locating survivors.
- To find stranded individuals in flooded areas by employing AI and image processing algorithms with thermal or visual cameras.
- To connect the two drones wirelessly so that they can share information and coordinate tasks in real time.
- To create an Internet of Things-based control system for mission updates, data logging, and remote monitoring.
- To utilize GPS-assisted navigation and optimize flight paths for the accurate and efficient delivery of alleviation substances.

1.5 Scope of the Project

The scope of this project focuses on the design and development of an autonomous dual UAV system utilizing a separate "scout-courier" architecture. The primary scout drone acts as an edge computing device and uses YOLOv5 deep learning algorithms to process information on a Raspberry Pi device to identify survivors in real time. Upon detection, the system uses a synchronized communication protocol to autonomously transmit precise geospatial coordinates to the secondary courier drone. The courier drone then navigates to the desired destination for the precise release of essential food and medical supplies. The project aims to provide a scalable and cost-effective solution without utilizing high-cost, industrial-grade military equipment.

1.6 Methodology Overview

The project methodology begins with research and problem identification regarding flood disaster scenarios and current UAV-based solutions. This is followed by the system design of two drones with distinct roles, including structural design and weight distribution analysis using CAD. Hardware development involves choosing and integrating components like GPS, wireless communication modules, flight control units, and microcontrollers. Software development includes the deployment of flight control logic, communication protocols for inter-drone coordination, and AI-based human detection algorithms utilizing OpenCV or TensorFlow. Finally, field exams and simulations are run to assess drone balance, detection accuracy, communication reliability, and reaction time.

1.7 Organization of the Report

This report is organized into five chapters: Chapter 1 introduces the project's scope and objectives; Chapter 2 reviews existing literature; Chapter 3 details the system design and methodology; Chapter 4 presents the implementation and results; and Chapter 5 concludes with future enhancements.

CHAPTER 2

LITERATURE REVIEW

Rathod, Puslekar, and Polaswar deployed CNN and YOLOv8 models on edge-equipped drones for real-time target identification in agricultural settings [1]. The system successfully achieved precise detection across varying lighting conditions, validating the feasibility of real-time aerial edge inference [1]. While its primary limitation is the contextual focus on precision agriculture rather than emergency response, its strength lies in optimizing YOLO architectures for edge devices [1]. Technically, this foundational work establishes the core edge-deployment methodology utilized for the Scout UAV's real-time detection pipeline [1].

Gupta et al. investigated the deployment of AI-driven UAVs specifically tailored for active flood monitoring and rescue mapping [2]. Their findings demonstrated that aerial AI significantly accelerates the mapping of inundated zones despite harsh environmental constraints [2]. The study's major strength is its direct focus on the environmental challenges of floods, though it is limited by the lack of a physical aid delivery mechanism [2]. Compared to generalized monitoring, this paper provides the critical environmental context that justifies a decoupled, multi-drone rescue architecture [2].

Nguyen et al. addressed the specific challenge of detecting human victims in turbulent water using specialized UAV vision algorithms [3]. The research successfully distinguished survivors from dynamic waves and floating debris, achieving high detection rates in complex aquatic environments [3]. The strength of this work is its robust optical filtering, while its limitation is the high computational intensity required for processing fluid backgrounds [3]. This paper is crucial as it provides the specialized visual filtering logic necessary for differentiating human targets from post-disaster water clutter [3].

B. Zhao et al. engineered a rapid UAV image processing pipeline designed for immediate flood target acquisition and emergency response [4]. The system effectively searched for people in flooded areas with minimal latency and significantly reduced false alarms [4]. Its core strength is the highly optimized aerial image pipeline for edge devices, though it lacks a secondary physical actuation mechanism for delivering aid [4]. This study directly inspires the proposed reconnaissance methodology, highlighting the necessity of a secondary "Courier" node to bridge visual identification and physical rescue [4].

Ulmamei et al. explored human detection by processing UAV thermal imagery entirely on resource-constrained embedded platforms like the Raspberry Pi [5]. The study demonstrated that microcomputers can effectively locate survivors during night operations or through dense disaster debris [5]. The major strength is its capability to operate in low-light conditions, while the limitation is the added weight and power draw of thermal sensors on the UAV payload [5]. This paper validates the use of embedded hardware for demanding inference loads, supporting the Scout UAV's computational design [5].

Shastri et al. investigated the implementation of Edge AI for drone-based search and rescue missions to bypass cloud reliance [6]. The authors proved that running detection models locally drastically reduces mission latency and generates faster alerts during SAR sweeps [6]. The primary strength is its complete independence from vulnerable cellular networks, though edge computing inherently restricts the size and depth of deployable neural networks [6]. This paper fundamentally justifies the Scout drone's edge-computing architecture, prioritizing localized data processing over high-latency cloud dependence [6].

Betti et al. developed a theoretical framework for multi-UAV task allocation and coordinate sharing in complex SAR operations [7]. The research modeled how multiple drones can efficiently divide search grids and share critical telemetry data to optimize rescue timelines [7]. The major strength is its robust algorithmic approach to workload splitting, though it is limited by a lack of hardware-level payload actuation metrics [7]. This study supports the core cyber-physical logic of the dual-drone system, validating the hand-off of GPS coordinates from the Scout to the Courier drone [7].

Ghamry et al. addressed the complex coordination and pathfinding required for multiple UAVs operating simultaneously in shared airspaces [8]. The authors developed algorithms to ensure collision avoidance and synchronized flight paths for decentralized multi-agent systems [8]. The core strength is its robust navigational safety protocols, while the limitation is the high software complexity required for implementation without a heavy ground station [8]. This paper backs up the critical navigational logic and failsafes required when both UAVs are airborne in the same disaster zone [8].

Sudhakar et al. detailed the design of an autonomous UAV engineered exclusively for safe medical supply delivery [9]. The research covered the mechanical and logistical requirements for carrying, stabilizing, and dropping payloads to isolated targets [9]. The major strength of this work is its deep focus on electromechanical release mechanisms, while its limitation is the assumption that target coordinates are known prior to launch [9]. This study directly informs the payload capacity constraints and the actuation system implemented on the proposed Courier drone [9].

Rabah et al. explored autonomous navigation protocols specific to precise aid delivery drops from UAVs [10]. The study mathematically modeled the precise flight maneuvers required for a UAV to safely descend, hover, drop an aid package, and execute a return trajectory [10]. The strength of this research is its precise altitude modeling, while its limitation is the system's susceptibility to localized wind shear during the descent phase [10]. This paper outlines the essential PID stabilization and altitude control logic the Courier UAV utilizes for precise 6-meter drops [10].

• **Summary of Literature Review :**

Ref.	Authors (Year)	Major Findings / New Work	Strengths & Limitations	Technical Comparison
[1]	Rathod et al. (2025)	Deployed AI models on edge hardware for real-time target identification.	Strength: Validates edge AI deployment. Limitation: Focuses on agricultural targets.	Establishes the foundational AI and edge-deployment methodology utilized for the Scout UAV.
[2]	Gupta et al. (2024)	Evaluated AI-driven UAVs specifically for active flood monitoring and mapping.	Strength: Addresses harsh flood environments. Limitation: Lacks a payload delivery mechanism.	Sets the environmental context that necessitates a decoupled, multi-drone architecture.
[3]	Nguyen et al. (2021)	Developed vision algorithms for detecting victims in turbulent water.	Strength: High detection in dynamic fluids. Limitation: Computationally intensive.	Provides the optical filtering logic necessary to distinguish humans from water and debris.
[4]	B. Zhao et al. (2024)	Engineered a rapid UAV image pipeline for immediate flood target acquisition.	Strength: Optimized for quick aerial inference. Limitation: No physical actuation phase.	Inspires the reconnaissance methodology, highlighting the need for a secondary Courier node.
[5]	Ulmamei et al. (2025)	Processed UAV thermal imagery entirely on resource-constrained platforms.	Strength: Enables night/low-light operations. Limitation: Thermal sensors add payload weight.	Proves that embedded hardware (Raspberry Pi) can reliably handle complex detection tasks.
[6]	Shastry et al. (2023)	Demonstrated that edge AI reduces SAR mission latency compared to cloud networks.	Strength: Highly reliable during network failures. Limitation: Restricts CNN model size/depth.	Justifies running detection models locally on the drone to bypass compromised cellular networks.

[7]	Betti et al. (2022)	Modeled multi-UAV task allocation and coordinate sharing for SAR operations.	Strength: Efficient multi-node workload splitting. Limitation: Primarily a theoretical framework.	Supports the core logic of the Scout drone communicating GPS coordinates to the Courier.
[8]	Ghamry et al. (2021)	Developed algorithms for collision avoidance and pathfinding in multi-UAV setups.	Strength: Robust shared-airspace safety protocols. Limitation: Complex software implementation.	Backs up the navigational safety logic required when both UAVs operate simultaneously.
[9]	Sudhakar et al. (2021)	Designed an autonomous UAV focused heavily on mechanical medical supply delivery.	Strength: Details payload logistics and release. Limitation: Assumes target coordinates are known.	Informs the mechanical release mechanism and payload capacity constraints of the Courier drone.
[10]	Rabah et al. (2022)	Addressed specific flight maneuvers for safe descent, hovering, and accurate aid drops.	Strength: Precise altitude and descent modeling. Limitation: Susceptible to descent wind shear.	Outlines the PID stabilization and altitude control logic the Courier uses for precise 6m drops.

CHAPTER 3

METHODOLOGY

3.1 Method Adopted to Solve the Problem

To address the severe operational limitations of traditional single-drone platforms specifically the trade-off between high-compute AI processing and heavy-lift battery endurance the method adopted for this project is a decentralized, heterogeneous dual-UAV architecture. The system introduces a specialized "Scout-Courier" framework designed to operate in GPS-denied or communication-degraded flood environments. By explicitly separating the aerial reconnaissance phase (survivor detection) from the physical logistics phase (aid delivery), the adopted method significantly optimizes battery consumption, flight endurance, and overall response latency.

3.2 Overview of Project Execution

The project is executed as a coordinated, closed-loop cyber-physical system where intelligence and physical actuation are distributed across three primary interconnected nodes:

- **The Scout UAV (Reconnaissance Node):** Serving as the "eyes" of the operation, this primary drone is dedicated exclusively to high-altitude aerial surveillance. It utilizes an onboard Raspberry Pi 5 to perform edge computing, running a continuous YOLOv5 detection pipeline to identify survivors without relying on cloud-based processing.
- **Ground Control Station (Centralized Routing Node):** Functioning as the computational "brain," the Ground Control Station (GCS) handles continuous telemetry routing and mission logic. It intercepts targeting data from the Scout UAV via a Skydroid T12 data link, validates the coordinates, and autonomously compiles a dispatch mission.
- **The Carrier UAV (Actuation Node):** Operating as the physical "hands," this secondary drone is engineered specifically for heavy-lift navigation. Upon receiving the automated dispatch command, it navigates to the validated coordinates to execute a precise electromechanical release of medical and food supplies.

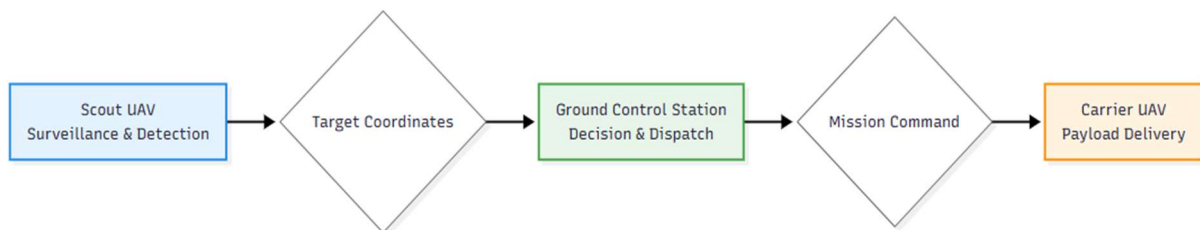


Fig. 1. Basic block diagram of telemetry data flow between the Scout, GCS, and Carrier nodes.

3.3 Step-wise Approach to the Solution

The operational workflow of the dual-drone system is strictly governed by a finite state machine that coordinates the tasks sequentially. The step-wise approach to executing the rescue mission is as follows:

Step 1: Mission Initialization and Autonomous Patrolling

The mission begins with the GCS uploading a predefined WGS84 search grid (GPS waypoints) to the Flight Control Unit (FCU) of the Scout drone. Upon engagement, the Scout autonomously navigates these waypoints while continuously processing aerial imagery through its real-time YOLOv5 vision pipeline.

Step 2: Real-Time Survivor Detection via Edge AI

As the Scout patrols, the YOLOv5 convolutional neural network processes RGB image arrays to extract high-level feature maps. The prediction head generates bounding boxes and assigns a confidence score (C) mathematically quantifying the certainty of a survivor's presence:

$$C = P(\text{Object}) \times \text{IoU}_{\text{truth}}^{\text{pred}}$$

To mitigate false positives caused by floating debris or water reflections, an empirical threshold is applied. The algorithm strictly rejects any prediction where the class-specific confidence score is less than 0.70 ($C_{\text{class}} < 0.70$).



Fig. 2. Graphical user interface of the detection system

Step 3: Geotagging and Coordinate Mapping

Once a survivor is positively identified, the Scout transitions into a stabilized "Loiter" flight mode. The system utilizes a geometric projection model that takes the UAV's current GPS location, barometric altitude, and the camera's Field of View (FOV) to translate the 2D pixel detection into a physical ground distance. This generates the estimated latitude and longitude of the target lat_{est} , lon_{est} .

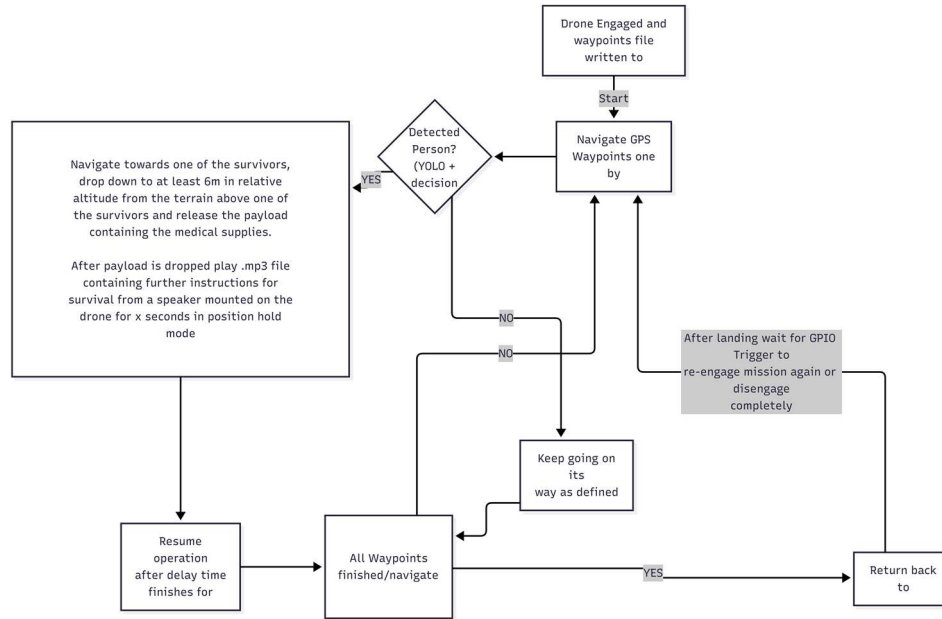


Fig. 3. Automated Scout-Courier Process Pipeline

Fig. 3. explains the decision-making workflow of the scout UAV during search operations. The drone navigates through predefined GPS waypoints while continuously performing YOLO-based human detection. If a person is detected, the drone descends to about 6 m above the terrain and releases a payload containing medical supplies. After dropping the payload, the drone plays an audio instruction (.mp3) to guide the survivor and temporarily holds position. Once the delay period ends, the drone resumes waypoint navigation until all waypoints are completed or returns based on mission logic.

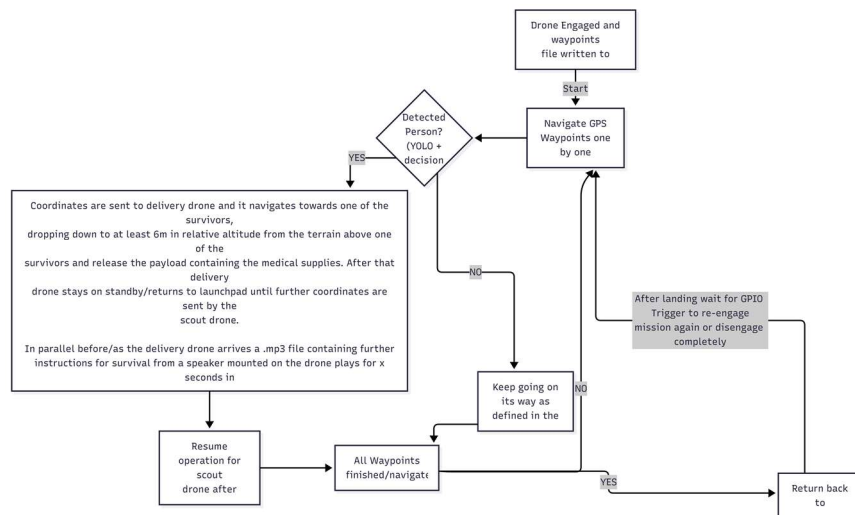


Fig. 4. Navigation Flowchart for GPS Target Computation

Fig. 4. describes the cooperative workflow between the scout drone and the delivery drone. The scout UAV scans waypoints and detects survivors using YOLO-based vision processing. When a survivor is detected, the coordinates are transmitted to the delivery drone, which navigates to the target location, descends to a safe altitude, and drops medical supplies. Simultaneously, the drone plays audio survival instructions through an onboard speaker. After completing the delivery, the drone either returns to the launch point or waits for new coordinates from the scout UAV to continue the rescue mission.

Step 4: Automated Handshake and Telemetry Routing

The Raspberry Pi 5 packages the generated GPS target coordinates into discrete MAVLink messages. These are broadcast via the Skydroid T12 transmitter to the GCS. An autonomous dispatch script on the GCS intercepts the packet, evaluates the data, and if the coordinates fall within the operational radius, instantly uploads a new waypoint mission to the Carrier UAV, triggering physical dispatch without human intervention.

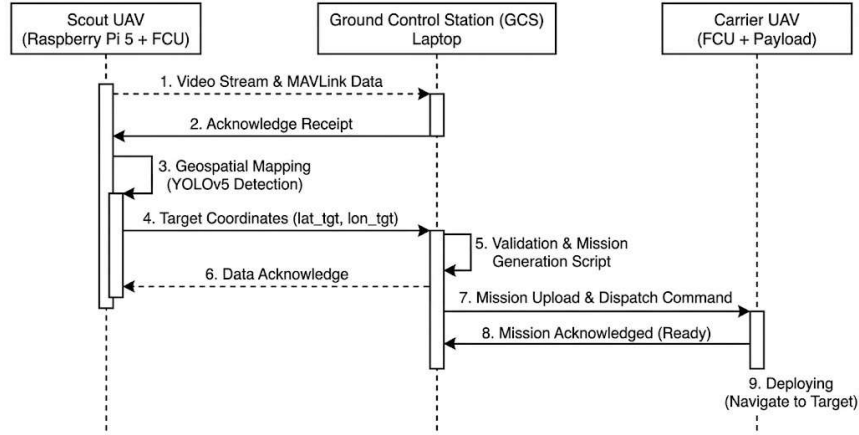


Fig. 5. Telemetry data flow and autonomous dispatch sequence.

Step 5: Coordinated Dispatch and Precision Navigation

Upon receiving the dispatch command, the Carrier UAV utilizes its on-board GNSS module to traverse to the target coordinates at a safe cruising altitude. Once the 2D positional threshold is reached, the flight controller initiates a controlled vertical descent, dropping to a relative altitude of 6 meters above the terrain to minimize the kinetic impact of the payload.

Step 6: Flight Stabilization, Payload Release, and Failsafe

During the critical descent phase, the Carrier UAV regulates its attitude and altitude against unpredictable wind gusts using a closed-loop Proportional-Integral-Derivative (PID) controller. The control signal $u(t)$ sent to the motors is defined as:

$$u(t) = K_p e(t) + K_i \int_{\{0\}}^{\{t\}} e(\tau) d\tau + K_d \frac{de(t)}{d(t)}$$

Once spatial stabilization is achieved at the 6-meter altitude, the FCU actuates a PWM-driven micro-servo to mechanically release the medical payload. Following the drop, both the Carrier and the Scout UAV trigger failsafe Return to Launch (RTL) commands, automatically returning to the launchpad to conclude the mission.

CHAPTER 4

Hardware/Software Design

4.1 Application Block Diagram and System Architecture The proposed disaster response framework is a closed-loop, coordinated cyber-physical system divided into three primary interconnected nodes: The Scout UAV, the Ground Control Station (GCS), and the Carrier UAV.

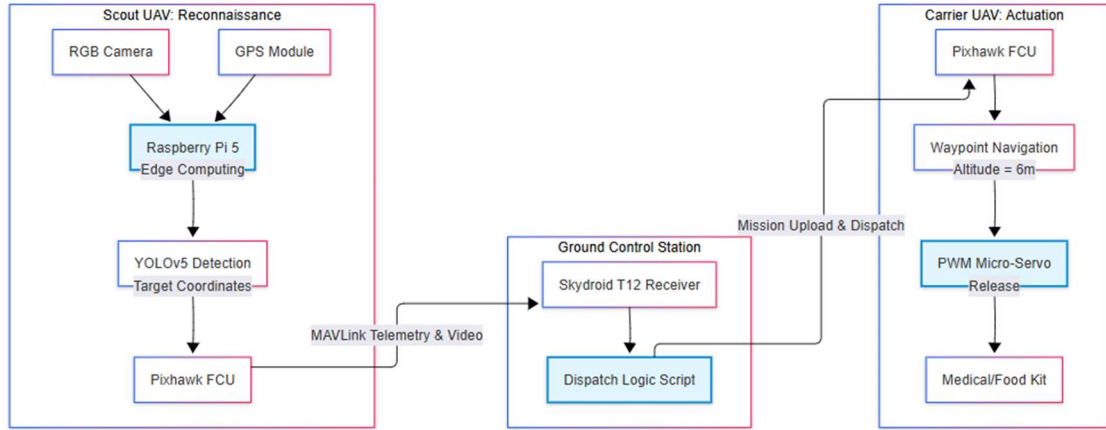


Fig. 6. High-level block diagram of the proposed cyber-physical system architecture, illustrating the hardware distribution and telemetry data flow between the Scout, GCS, and Carrier nodes

The system operates on a decoupled "Scout-Courier" architecture to overcome the battery endurance limitations of single heavy-lift drones. The components making up this block diagram include:

- **The Scout UAV (Node 1):** Dedicated to high-altitude aerial reconnaissance. It houses the edge AI processing unit to run visual detection algorithms independently.
- **The Carrier UAV (Node 2):** Dedicated to heavy-lift physical logistics. It houses the electromechanical actuation system to drop the medical/food payload.
- **Ground Control Station (Node 3):** Acts as the centralized routing hub that intercepts telemetry data from the Scout and autonomously dispatches the Carrier.

Component	Scout UAV (Node 1)	Carrier UAV (Node 3)
Chassis	F450 Quadcopter	F450 Quadcopter
Flight Controller	SpeedyBee F405 V4 BLS 55A 30x30 Stack	Pixhawk (PX4/ArduPilot)
Companion PC	Raspberry Pi 5 (Edge AI)	None (Relies on FCU)
Primary Payload	RGB Camera	Medical/Food Kit
Actuation	None	PWM Micro-Servo
Telemetry	Skydroid T12 (Video+Data)	Skydroid T12 (Data only)

Table 2. Hardware Specifications and Payload Distribution

4.2 Hardware Subsystem Design and Implementation

To ensure modularity and maintainability, both UAVs are built on a standardized F450 quadcopter chassis. High KV brushless DC motors are driven by a centralized 4-in-1 Electronic Speed Controller (ESC) connected directly to the main Flight Control Unit (FCU).

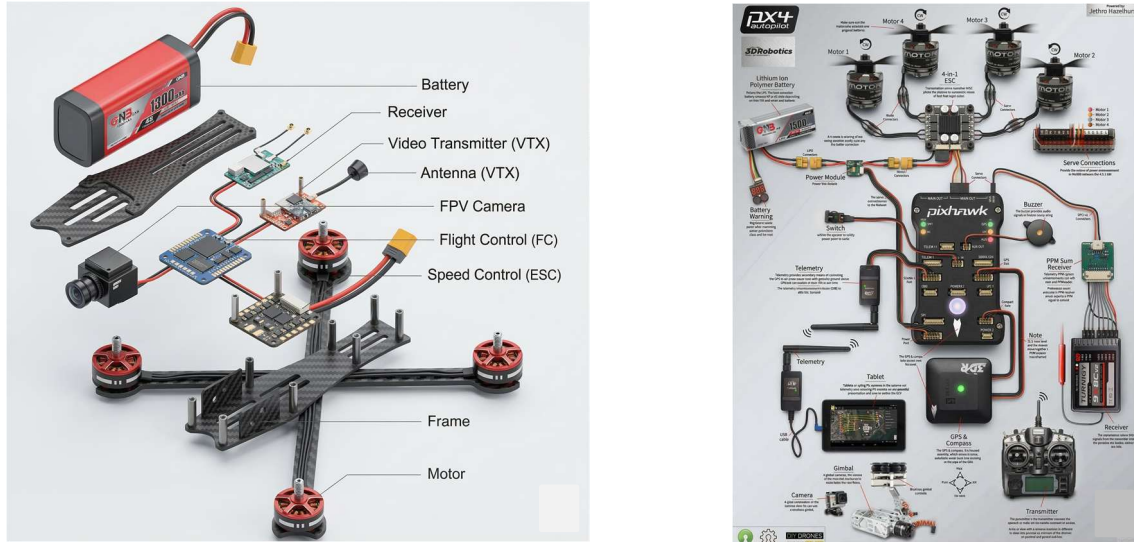


Fig. 7. Comprehensive hardware integration diagram detailing the flight controller, 4-in-1 ESC, propulsion mapping, and telemetry layout

- **Flight Controllers and Navigation:** Both drones utilize the Pixhawk (PX4/ArduPilot) FCU. This controller is equipped with a GNSS unit for spatial tracking, an Inertial Measurement Unit (IMU) for orientation and balance, and a barometer for measuring altitude.
- **Edge Computing Interface:** The Scout UAV is equipped with a Raspberry Pi 5 acting as a companion computer. The GPIO TX/RX pins of the Raspberry Pi are wired directly to the Telemetry 2 (Serial 2) UART port on the Pixhawk. This creates a two-way MAVLink communication bridge, allowing the Pi to inject real-time navigation commands into the flight loop.
- **Payload Actuation Mechanism:** The Carrier UAV features a centrally mounted payload mechanism to maintain the center of gravity. A high-torque PWM micro-servo is connected to the Pixhawk's auxiliary output channels. When the drone stabilizes at the 6-meter drop altitude, the FCU sends a pulse-width modulation (PWM) signal to open the servo and deploy the 0.5 to 1.0 kg survival kit.
- **Telemetry Data Link:** Both drones maintain a secure 2.4 GHz communication link with the GCS using the Skydroid T12 system. This dual-band transmitter handles both high-bandwidth video streams and low-bandwidth MAVLink telemetry packets, while also providing a hardware-level manual RC override switch for safety.



Fig. 8. Dual Drone System

4.3 Software Design and Implementation

The software architecture is distributed across perception, navigation, and communication modules.

- **Perception Module (Edge AI):** The Scout UAV uses Python and OpenCV to run a YOLOv5 convolutional neural network on the Raspberry Pi 5. The Cross Stage Partial Network (CSPDarkNet) backbone processes RGB frames to detect human heat signatures and visual profiles in near real-time. Predictions with a confidence score (C_{class}) below 0.70 are algorithmically rejected to mitigate false positives.
- **Geospatial Mapping:** When a bounding box is generated, the software uses the drone's instantaneous barometric altitude and camera field-of-view (FOV) to translate the 2D pixel centroid into physical WGS84 coordinates (lat_{est} , lon_{est}).

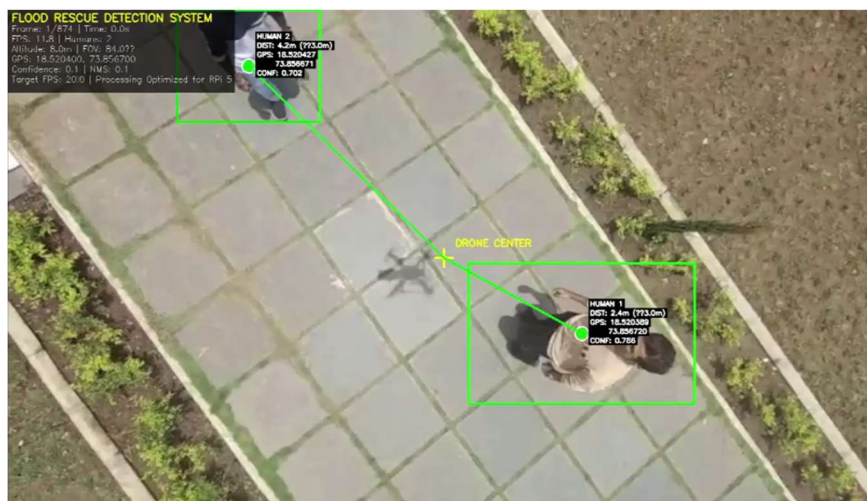


Fig. 9. Graphical user interface of the detection system illustrating the real time geometric projection from the Drone Center to the target, calculating relative distance and absolute GPS coordinates.

4.4 System Algorithms and State Machine

The autonomous workflow of the system is tightly controlled by a finite state machine that synchronizes the operation of both drones. The core logic ensuring the drones operate without manual human input is the Autonomous Dispatch Algorithm running on the GCS laptop.

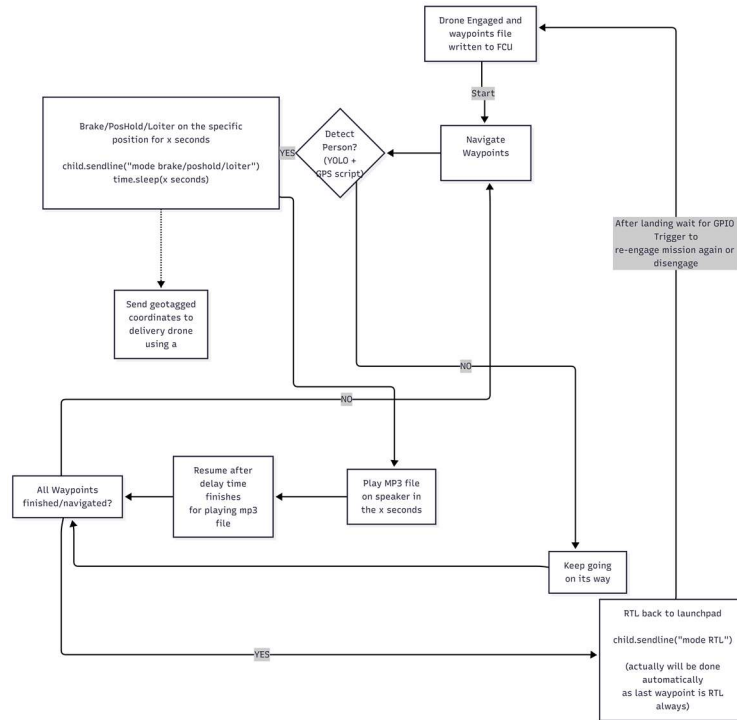


Fig. 10. Finite State Machine for Autonomous Backend Software Execution

Algorithm: Autonomous Dispatch Logic (GCS)

The following step-wise algorithm defines how the system transitions from target detection to physical payload dispatch:

Require: Video Stream (V), GPS Data (G) from Scout UAV

Ensure: Carrier Dispatch Command (D)

- 1: Initialize Skydroid T12 Telemetry Link
- 2: While Scout UAV is Airborne do
- 3: Run YOLOv5_Detect(V)
- 4: If Confidence_Threshold > 0.70 then
- 5: Extract_Coordinates(G) to generate Target_Lat, Target_Lon
- 6: Verify_Coordinates(Target_Lat, Target_Lon)
- 7: Transmit(Target_Lat, Target_Lon) to GCS via MAVLink
- 8: If GCS_Receive_Success == True then
- 9: Generate_Waypoint_Mission(Target_Lat, Target_Lon)
- 10: Upload_Mission_to_Carrier_UAV()
- 11: Execute_Dispatch(Carrier_UAV)
- 12: Break loop
- 13: End If
- 14: End If
- 15: End While

CHAPTER 5

Experimental Results

5.1 Experimental Setup

To rigorously test the performance, reliability, and advantages of the proposed Scout-Courier dual UAV system, extensive experiments were conducted within simulated flood scenarios. The primary objective of these field trials was to evaluate the effectiveness of the AI detection system, measure the latency of the communication network, and assess the success rate of the physical payload delivery across various environmental parameters.

5.2 Quantitative Results

Methodology	Detection Accuracy	Coordination Speed	Human Intervention
Manual SAR	78.3%	10-15 Minutes	High
Single Drone	85.2%	1.2 Minutes	Moderate
Proposed Dual-Drone	92.5%	1.2 Seconds	Minimal

Table 3. Comparison of Rescue Efficiency and Coordination

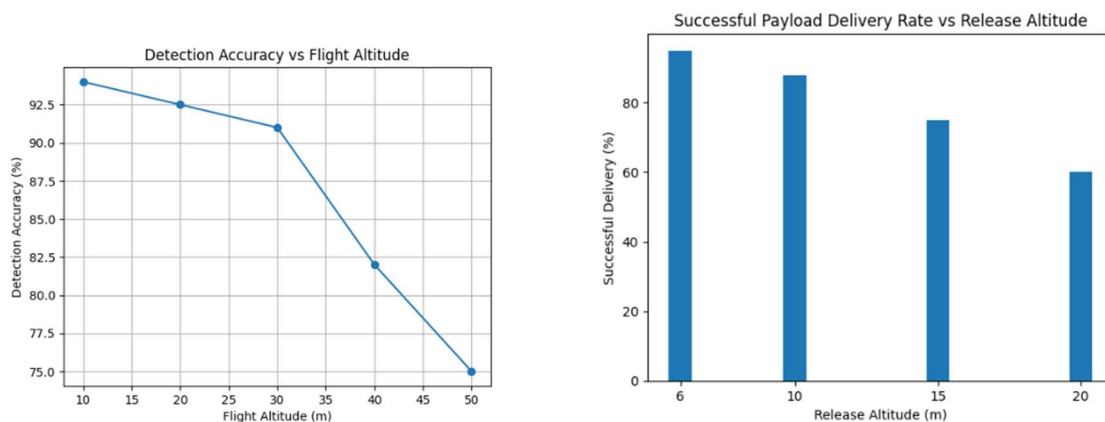


Fig. 11. Detection Accuracy Vs Flight Altitude. Fig. 12. Payload Delivery Vs Release Altitude

The graph in Fig. 11 demonstrates that the scout UAV's detection accuracy is inversely proportional to its flight altitude, peaking near 94% at 10 meters and dropping significantly to 75% at 50 meters. Similarly, the bar graph in Fig. 12 highlights that the carrier UAV's payload delivery success rate is also heavily dependent on its release height. Optimal delivery precision is achieved at a lower altitude of 6 meters (exceeding 90% success), while dropping the payload from 20 meters causes the success rate to fall drastically to around 60%. Together, these graphs validate that operating at strict, lower altitudes is critical for the system to ensure both reliable survivor detection and accurate aid delivery.

5.3 Evaluation of Results

The quantitative results demonstrate highly successful operational metrics across both the reconnaissance and logistics phases. The edge-computed YOLOv5 vision system achieved a baseline detection accuracy of 92.5% with a competitively low false positive rate (FPR) of 7.5%. As illustrated in the altitude correlation data, the system's accuracy is heavily dependent on flight height; detection rates are optimal between 10 meters and 30 meters, but degrade rapidly for pixels captured above 40 meters.

In terms of system coordination and latency, the decentralized architecture achieved an average response time of 2.3 seconds from the moment of survivor detection to the coordinated handover. The Skydroid T12 data link was highly reliable, maintaining a stable 500ms delay during the transmission of geotagged MAVLink messages. Consequently, the dual-drone approach slashed the mission coordination time to just 1.2 seconds with minimal human intervention, vastly outperforming the 1.2 minutes required by single drone systems and the 10 to 15 minutes needed for manual search operations. Finally, physical payload delivery tests revealed that the carrier UAV achieves its highest success rate (over 90%) when hovering at an optimal relative altitude of 6 meters. However, payload delivery accuracy experienced severe degradation when dropped from higher altitudes of 15 meters to 20 meters.

5.4 Analysis of Obtained Results

The experimental findings decisively validate the superiority of the decoupled "Scout-Courier" architecture over traditional search and rescue (SAR) methodologies. The 92.5% detection accuracy confirms that running complex Convolutional Neural Networks (CNNs) directly on resource-constrained edge hardware (Raspberry Pi) is highly effective, entirely eliminating the latency associated with cloud computing dependencies. However, the observed degradation in detection accuracy at higher altitudes limits the Scout UAV's effective operational ceiling to 50 meters for high-resolution imaging. Furthermore, the system's susceptibility to dense vegetation, submerged debris, and dark environments indicates a need for adaptive filtering thresholds and the future integration of infrared cameras for nocturnal operations.

The minimal 1.2-second coordination time proves that the automated handshake protocol successfully bridges the gap between detection and physical intervention without bottlenecking the system. The sharp decline in delivery accuracy above 15 meters highlights the profound impact of parabolic trajectory drift and wind shear, fully justifying the system's programmed descent to a strict 6-meter threshold prior to payload release. To further mitigate environmental constraints such as extreme wind turbulence which causes motion blur and elevated false positive rates the dynamic restriction of the Scout UAV's cruising speed to a range of 3 m/s to 5 m/s proved to be a vital aerodynamic safeguard.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project successfully developed a novel decentralized dual-UAV approach to optimize autonomous search and rescue (SAR) missions in flood-affected areas. By splitting reconnaissance and logistics into a specialized "scout-courier" model, the system effectively overcomes the battery endurance issues of conventional SAR drones. Extensive field testing proved the system's high reliability, with the YOLOv5 edge-computed CNN model aboard the Scout UAV achieving a high accuracy level of 92.5% in detecting human targets. The automated handshake protocol between the drones significantly reduced response time, coordinating a dispatch in a lightning-fast 1.2 seconds. Logically, this rapid response was supported by a highly effective geometric projection model for geospatial mapping, which resulted in a low positional error rate of only 10.95 meters. This precise targeting allowed the Carrier UAV to deliver the payload to the target location with a high success rate of over 90% when dropped at an experimentally determined optimal height of 6 meters.

6.2 Future Scope

Although the current architecture achieves significant advances in the effectiveness of daylight SAR systems, next versions will emphasize expanding versatility. To overcome the rapid loss of accuracy of standard RGB camera systems in low light conditions, future iterations will include the fusion of thermal and IR sensor data in the edge calculation process. Additionally, the communications architecture will be expanded to include swarm-based coordination for multiple scout UAV systems to disperse and sweep large disaster areas.

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Appendix

Plagiarism Report



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Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



Project Title Mapping with POs and PSOs

Project Title: Design and Development of Autonomous Dual-Drone System for Survivor Detection and Aid Delivery in Flood Zones

Gr. No	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
1	3	3	3	3	3	2	2	3	2	3	2	3	3	3

Justification:

- **PO1 – Engineering expertise:** Basic ideas of electronics, conversation and manage systems had been used within the design of the twin drone device.
- **PO2 – Problem evaluation:** The demanding situations of flood region, detection accuracy and conversation range have been analyzed to outline system requirements.
- **PO3 – Design/development of answers:** Designed and evolved an independent drone pair able to finding survivors and efficaciously delivering resource.
- **PO4 – Investigating complex troubles:** Experiments and assessments were done on sensors, flight controls and photograph processing to verify the system's performance.
- **PO5 – Use of current equipment:** Modern tools inclusive of simulation software program, drone control interfaces and AI-based photograph processing have been used.
- **PO6 – Engineers and Society:** Social needs were met via offering rescue-orientated structures to help catastrophe management and human protection.
- **PO7 – Environment and sustainability:** Designed power-efficient drones that lessen the environmental effect during operation.
- **PO8 – Ethics:** Ethical issues have been ensured by means of selling drone programs which can be secure, responsible and recognize privateness.
- **PO9 – Individual and teamwork:** The venture become finished via powerful teamwork and coordination amongst participants in all levels.
- **PO10 – Communication:** Results and development were correctly communicated through reports, presentations and documentation.
- **PO11 – Project control and finance:** Resources and additives have been correctly managed whilst keeping timelines and value effectiveness.
- **PO12 – Lifelong mastering:** Encouraged to analyze new technology consisting of AI, IoT and drone navigation for non-stop improvement of abilities.
- **PSO1:** Basic knowledge of electronics and telecommunications structures used inside the development of self-sustaining drone platforms.
- **PSO2:** Software equipment, check gadget and dimension device are used efficiently to achieve correct gadget overall performance.



Department of Electronics and Telecommunication Engineering
ONE PAGE ABSTRACT

Design and Development of Autonomous Dual-Drone System for Survivor Detection and Aid Delivery in Flood Zones

SDG 11: Sustainable Cities and Communities

Division- B.Tech [B] **Group Number-** B-5

1. Nirmithi Puslekar (B-109)
2. Sumit Rathod (B-110)
3. Onkar Sapkal (B-116)

Name of Internal Guide – Prof. R.N.Jadhav

Problem Statement - Flood zones present extreme dangers and severe accessibility challenges that hinder manual rescue operations. There is an urgent need for an automated, dual-drone system capable of safely and rapidly navigating these hazardous environments to detect survivors and deliver immediate, life-saving aid.

Project Domain: Robotics and Automation

Objectives -

1. To design and develop an autonomous dual-drone system capable of safely navigating hazardous and inaccessible flood zones.
2. To implement an AI-powered image processing model for the real-time and highly accurate detection of stranded survivors.
3. To engineer a reliable payload drop mechanism for the safe and targeted delivery of essential aid to the located victims.

Abstract –

Floods are one of the maximum devastating natural disasters, often inflicting huge destruction, displacement and loss of existence. Rapid detection of survivors and timely transport of important help is critical to minimize injuries. This project offers the design and development of an autonomous twin drone system capable of locating survivors and presenting help in flood-affected regions. The system integrates two specialized drones: Drone A for aerial surveillance and Drone B for payload transport. Drone A is prepared with a thermal imaging digital camera and GPS module that autonomously scans flooded regions, detects human warmth signatures and marks particular survivor coordinates using photo processing and AI-based detection algorithms. These coordinates are then transmitted wirelessly to Drone B, which autonomously navigates to the region and promises emergency materials which include food, water and medical kits. Both drones communicate via a stable IoT-based control network for real-time tracking and coordination. This device focuses on autonomy, accuracy, and efficiency while reducing human intervention during rescue operations. Simulation and prototype studies prove that the proposed structure of a twin-drone could enhance response times, operational reliability, and overall performance in disaster control cases.

Project photo including Group members and guide during Presentation (Geo Tag Photo).



Project Mapping with Program Outcomes (1 – Slight, 2- Moderate, 3- Substantial)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
	3	3	3	3	3	2	2	3	2	3	2	3	3	3

Participation in Competition: National Innovation Challenge for Drone Application and Research

Awards (if any): AIR 24

Team



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Honorary Secretary

DR. PRADEEP B. MANE
Principal

Ref. No.: AISSMS/2021/2024-25/3682

Date: 16/04/2025

Team Details for NIDAR - National Innovation Challenge for Drone Application & Research Competition

Institute Name: All India Shri Shivaji Memorial Society's Institute of Information Technology, Pune.

Team Name: Aero Guardian

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1	Aryan Basnet	Electronics and Telecommunication Engineering	2024-25 SY B.Tech	XIIE2005	2561 4086 9987
2	Pushkar Lokhande	Electronics and Telecommunication Engineering	2024-25 SY B.Tech	XI-23034	3393 3889 6791
3	Sumit Rathod	Electronics and Telecommunication Engineering	2024-25 TY B.Tech	XI-22057	8957 2111 6536
4	Nirmiti Puslekar	Electronics and Telecommunication Engineering	2024-25 TY B.Tech	XI-22046	4603 3552 6294
5	Onkar Sapkal	Electronics and Telecommunication Engineering	2024-25 TY B.Tech	XI-22022	6495 2087 9389
6	Akash Meshram	Artificial Intelligence and Data Science	2024-25 SY B.Tech	AI-22056	2096 6533 9696
7	Anagha Chandan	Artificial Intelligence and Data Science	2024-25 SY B.Tech	AI-D2416	7843 6074 5925
8	Yash Yogesh Potdar	Artificial Intelligence and Data Science	2024-25 TY B.Tech	AI-22144	5086 7096 8477
9	Swaraj Ashok Chavan	Electronics and Telecommunication Engineering	2024-25 SY B.Tech	XI-23095	9911 8350 9789
10	Mr. Rahul Namdeo Jadhav	Electronics and Telecommunication Engineering	Faculty Coordinator	Faculty ID :	5751 0418 5903



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India's largest National Drone Innovation Challenge

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under the SwaYaan Initiative

Keep flying high!

Drone Club
Coordinators

Mr. R. N. Jadhav
Mr. G. H. Wani

Regards

Dr. P. B. Mane
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Department of Electronics and Telecommunication Engineering

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Project Domain:	Group No. :

Roll No.	Name	Email ID.	Mobile No.	Sign.

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Presentation 1 Evaluation

Criteria ↓	Student's Name	Student 1	Student 2	Student 3
	Sign			
Literature Review	(4 Marks)			
Formulation of Aim and Objectives	(4 Marks)			
Timely Submission	(2 Marks)			
Total Marks				

TW Submission Checklist

Synopsis (Three Different Copies)	
Log Book	
Project Phase 1 Report with Plagiarism Report (2 Copies)	
Project Title Mapping with POs and PSOs	
Presentation PPT	
Sponsorship Letter (if any)	

Presentation 2 Evaluation

Criteria ↓	Student's Name	Student 1	Student 2	Student 3
	Sign			
System Design	(4 Marks)			
Simulation using Modern Computer Aided Tools	(4 Marks)			
Timely Submission	(2 Marks)			
Total Marks				

Guide Sign.	
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Project Phase 2

Presentation 3 Evaluation

Criteria ↓	Student's Name	Student 1	Student 2	Student 3
	Sign			
System Implementation	(4 Marks)			
Working Model	(4 Marks)			
Timely Submission	(2 Marks)			
Total Marks				

TW Submission Checklist

Project Abstract	
Plagiarism Report	
Published Paper in UGC/WoS/Scopus care Journal with DOI	
Two Signed copies of report with CD	
Participation Certificate	
Project Hardware	
Project group photo with Guide	

Presentation 4 Evaluation

Criteria ↓	Student's Name	Student 1	Student 2	Student 3
	Sign			
Working Model with Enclosure	(3 Marks)			
Effective Presentation	(3 Marks)			
Project Report	(3 Marks)			
Timely Submission	(1 Marks)			
Total Marks				

Guide Sign.	
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Paper Title:		Journal/Conference Name:	
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Department of Electronics and Telecommunication Engineering
B. Tech Project
Academic Year: 2025-26
Self-Evaluation Sheet 1

- **Group Number: B-5**
- **Project Title: Design and Development of Autonomous Dual-Drone System for Survivor Detection and Aid Delivery in Flood Zones**
- **Name of Students:**
 1. Nirmity Puslekar
 2. Sumit Rathod
 3. Onkar Sapkal

Literature Survey	Design	Implementation	Test Result	Attendance According to Plan Activity	Working According to Plan Activity	Maintaining Log Book	Research Paper	Participation in Project Competition	Award/ Price if any
(5)	(20)	(20)	(5)	(10)	(5)	(5)	(5)	(5)	(5)

Guide Name: Prof. R. N. Jadhav

Sign



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Department of Electronics and Telecommunication Engineering

B. Tech Project

Academic Year: 2025-26

Self-Evaluation Sheet 2

Group Number	B-5
Project Title	Design and Development of Autonomous Dual-Drone System for Survivor Detection and Aid Delivery in Flood Zones
Date	18/05/2026

- 1. Describe your feelings about working on the project. Did you enjoy the working on it?**
I found building this humanitarian-focused project highly engaging and deeply rewarding as it perfectly bridged my ENTC and AI skills.
- 2. What was the hardest part about working on your project?**
The hardest part was optimizing the YOLOv5 computer vision model to run accurately in real-time on edge devices like the Raspberry Pi.
- 3. List some of the things you learnt while working on your project?**
I learned edge AI deployment, advanced ESP32 and Raspberry Pi communication, and hardware-software interfacing for payload delivery.
- 4. Were you satisfied with your final project?**
Yes, I am highly satisfied because our functional prototype successfully proved the viability of an autonomous "detect and sustain" system.
- 5. What did you like the best about your project?**
The most rewarding part was successfully executing the autonomous handoff where the AI detection perfectly triggered the physical payload drop.
- 6. What did you like the least about the project?**
The most frustrating part was dealing with unpredictable physical variables like battery drain, wind interference, and Wi-Fi signal drops.
- 7. What was the most important thing that you learnt from doing your project that will help you in the future?**
The most critical takeaway was mastering systems architecture by successfully integrating machine learning, networking protocols, and mechanical hardware.

Student Name

Sign

1. Nirmiti Puslekar

2. Sumit Rathod

3. Onkar Sapkal

Guide Name: Prof. R. N. Jadhav

Sign



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Department of Electronics and Telecommunication Engineering

B. Tech Project

Academic Year: 2025-26

Critical Thinking Questionnaire

Group Number	B-5
Project Title	Design and Development of Autonomous Dual-Drone System for Survivor Detection and Aid Delivery in Flood Zones
Date	18/05/2026

1. Who benefits from the project?

Disaster survivors needing immediate aid and search and rescue teams requiring rapid situational awareness benefit the most.

2. Who is this project harmful to?

It poses minimal harm, though there are potential secondary risk regarding citizen privacy from cameras and accidental drone collisions.

3. What are the strengths and weaknesses of the project?

The strength is rapid, targeted aid delivery in inaccessible areas, while the weakness is its heavy reliance on limited battery life and clear weather.

4. Where would we see this project in real world?

It would be deployed by emergency responders and incident command centers in disaster struck regions like flood zones or earthquake epicenters.

5. When would this project benefit our society?

It provides the highest benefit during the critical “golden hours” immediately following a natural disaster when rapid intervention is needed.

6. Why is this project a problem/challenge?

Coordinating autonomous dual flight, live AI video processing, and heavy payload delivery simultaneously is technically complex and resource intensive.

7. How does this project benefit us/others?

It saves lives by autonomously dropping essential survival kits to stranded victims long before human rescue teams can physically reach them.

8. Where are the areas for improvement?

Future iterations could improve by adding thermal imaging for night time detection and integrating long range communication protocols for off grid operation.

Student Name

Sign

1. Nirmity Puslekar

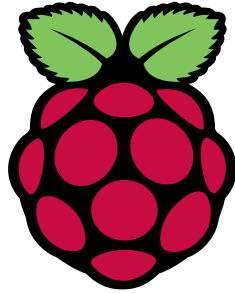
2. Sumit Rathod

3. Onkar Sapkal

Guide Name: Prof. R. N. Jadhav

Sign

DATASHEET



Raspberry Pi 4 Model B

Release 1.1

March 2024

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Table 1: Release History

Release	Date	Description
1	21/06/2019	First release
1.1	12/03/2024	Updated obsolescence statement and electrical specification

The latest release of this document can be found at <https://www.raspberrypi.org>



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1 Introduction

The Raspberry Pi 4 Model B (Pi4B) is the first of a new generation of Raspberry Pi computers supporting more RAM and with significantly enhanced CPU, GPU and I/O performance in a similar form factor, power envelope and cost as the previous generation Raspberry Pi 3B+.

The Pi4B is available with 1, 2, 4, or 8 gigabytes of LPDDR4 SDRAM.



2 Features

2.1 Hardware

- Quad core 64-bit ARM-Cortex A72 running at 1.5GHz
- 1, 2 and 4 Gigabyte LPDDR4 RAM options
- H.265 (HEVC) hardware decode (up to 4Kp60)
- H.264 hardware decode (up to 1080p60)
- VideoCore VI 3D Graphics
- Supports dual HDMI display output up to 4Kp60

2.2 Interfaces

- 802.11 b/g/n/ac Wireless LAN
- Bluetooth 5.0 with BLE
- 1x SD Card
- 2x micro-HDMI ports supporting dual displays up to 4Kp60 resolution
- 2x USB2 ports
- 2x USB3 ports
- 1x Gigabit Ethernet port (supports PoE with add-on PoE HAT)
- 1x Raspberry Pi camera port (2-lane MIPI CSI)
- 1x Raspberry Pi display port (2-lane MIPI DSI)
- 28x user GPIO supporting various interface options:
 - Up to 6x UART
 - Up to 6x I2C
 - Up to 5x SPI
 - 1x SDIO interface
 - 1x DPI (Parallel RGB Display)
 - 1x PCM
 - Up to 2x PWM channels
 - Up to 3x GPCLK outputs

2.3 Software

- ARMv8 Instruction Set
- Mature Linux software stack
- Actively developed and maintained



- Recent Linux kernel support
- Many drivers upstreamed
- Stable and well supported userland
- Availability of GPU functions using standard APIs

3 Mechanical Specification

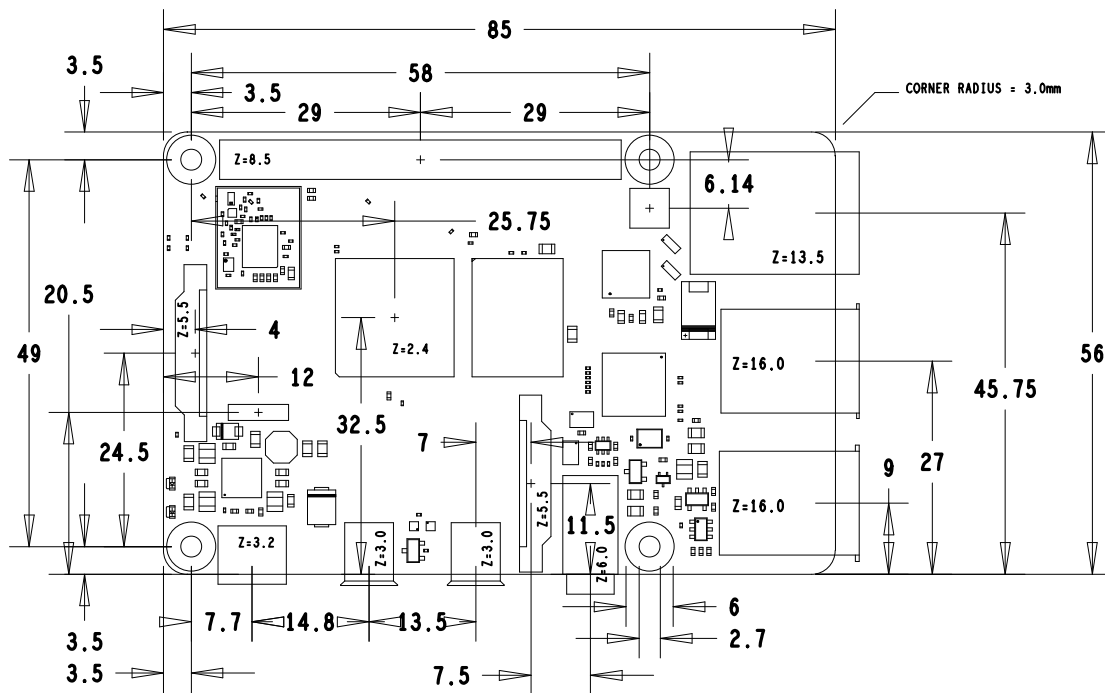


Figure 1: Mechanical Dimensions

4 Electrical Specification

Caution! Stresses above those listed in Table 2 may cause permanent damage to the device. This is a stress rating only; functional operation of the device under these or any other conditions above those listed in the operational sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Symbol	Parameter	Minimum	Maximum	Unit
VIN	5V Input Voltage	-0.5	6.0	V

Table 2: Absolute Maximum Ratings

Please note that VDD_IO is the GPIO bank voltage which is tied to the on-board 3.3V supply rail.



Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Unit
V_{IL}	Input low voltage ^a	VDD_IO = 3.3V	0	-	0.8	V
V_{IH}	Input high voltage ^a	VDD_IO = 3.3V	2.0	-	VDD_IO	V
I_{IL}	Input leakage current	TA = +85°C	-	-	10	μA
C_{IN}	Input capacitance	-	-	5	-	pF
V_{OL}	Output low voltage ^b	VDD_IO = 3.3V, IOL = -2mA	-	-	0.4	V
V_{OH}	Output high voltage ^b	VDD_IO = 3.3V, IOH = 2mA	VDD_IO - 0.4	-	-	V
I_{OL}	Output low current ^c	VDD_IO = 3.3V, VO = 0.4V	7	-	-	mA
I_{OH}	Output high current ^c	VDD_IO = 3.3V, VO = 2.3V	7	-	-	mA
R_{PU}	Pullup resistor	-	18	47	73	kΩ
R_{PD}	Pulldown resistor	-	18	47	73	kΩ

^a Hysteresis enabled

^b Default drive strength (8mA)

^c Maximum drive strength (16mA)

Table 3: DC Characteristics

Pin Name	Symbol	Parameter	Minimum	Typical	Maximum	Unit
Digital outputs	t_{rise}	10-90% rise time ^a	-	TBD	-	ns
Digital outputs	t_{fall}	90-10% fall time ^a	-	TBD	-	ns

^a Default drive strength, CL = 5pF, VDD_IO = 3.3V

Table 4: Digital I/O Pin AC Characteristics

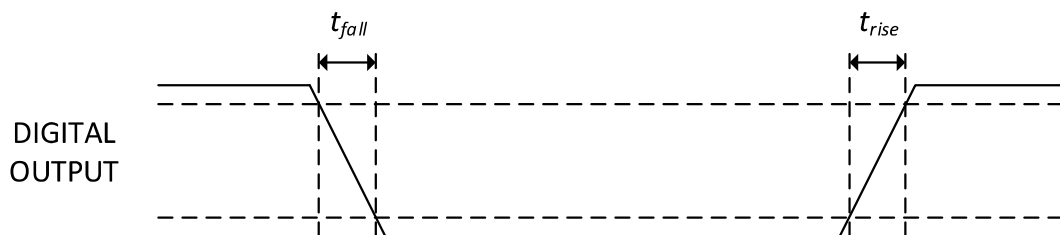


Figure 2: Digital IO Characteristics

4.1 Power Requirements

The Pi4B requires a good quality USB-C power supply capable of delivering 5V at 3A. If attached downstream USB devices consume less than 500mA, a 5V, 2.5A supply may be used.



5 Peripherals

5.1 GPIO Interface

The Pi4B makes 28 BCM2711 GPIOs available via a standard Raspberry Pi 40-pin header. This header is backwards compatible with all previous Raspberry Pi boards with a 40-way header.

5.1.1 GPIO Pin Assignments

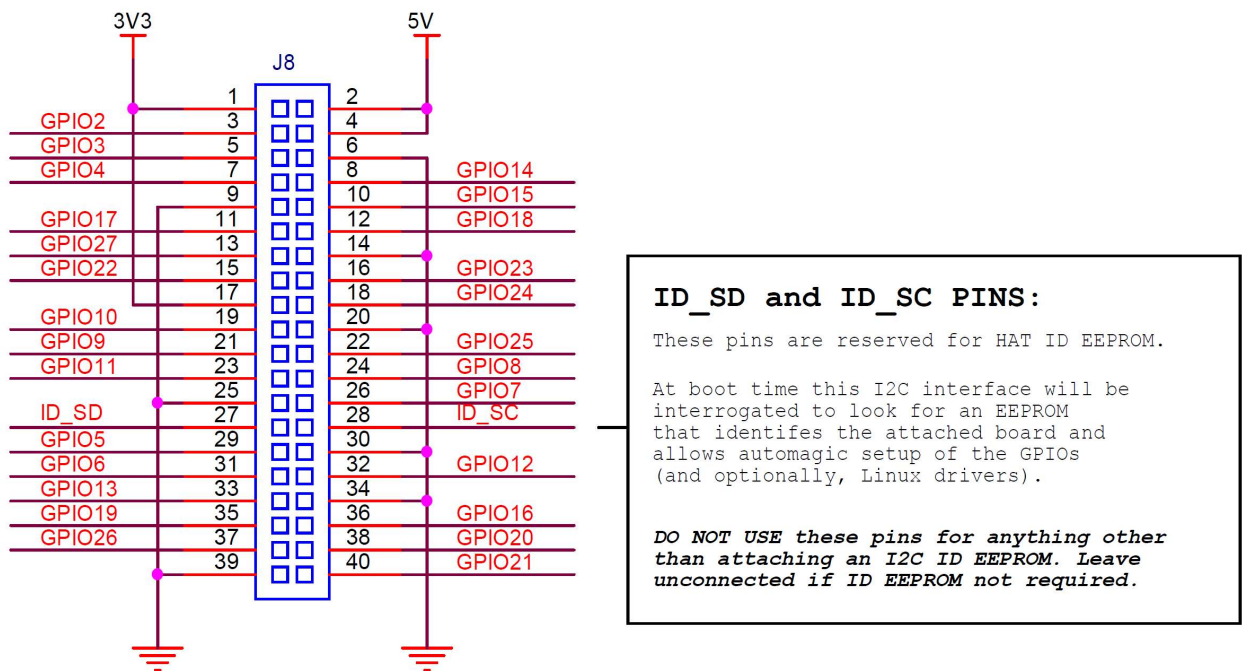


Figure 3: GPIO Connector Pinout

As well as being able to be used as straightforward software controlled input and output (with programmable pulls), GPIO pins can be switched (multiplexed) into various other modes backed by dedicated peripheral blocks such as I2C, UART and SPI.

In addition to the standard peripheral options found on legacy Pis, extra I2C, UART and SPI peripherals have been added to the BCM2711 chip and are available as further mux options on the Pi 4. This gives users much more flexibility when attaching add-on hardware as compared to older models.



5.1.2 GPIO Alternate Functions

GPIO	Default Pull	ALT0	ALT1	ALT2	ALT3	ALT4	ALT5
0	High	SDA0	SA5	PCLK	SPI3_CE0_N	TXD2	SDA6
1	High	SCL0	SA4	DE	SPI3_MISO	RXD2	SCL6
2	High	SDA1	SA3	LCD_VSYNC	SPI3_MOSI	CTS2	SDA3
3	High	SCL1	SA2	LCD_HSYNC	SPI3_SCLK	RTS2	SCL3
4	High	GPCLK0	SA1	DPI_D0	SPI4_CE0_N	TXD3	SDA3
5	High	GPCLK1	SA0	DPI_D1	SPI4_MISO	RXD3	SCL3
6	High	GPCLK2	SOE_N	DPI_D2	SPI4_MOSI	CTS3	SDA4
7	High	SPI0_CE1_N	SWE_N	DPI_D3	SPI4_SCLK	RTS3	SCL4
8	High	SPI0_CE0_N	SD0	DPI_D4	-	TXD4	SDA4
9	Low	SPI0_MISO	SD1	DPI_D5	-	RXD4	SCL4
10	Low	SPI0_MOSI	SD2	DPI_D6	-	CTS4	SDA5
11	Low	SPI0_SCLK	SD3	DPI_D7	-	RTS4	SCL5
12	Low	PWM0	SD4	DPI_D8	SPI5_CE0_N	TXD5	SDA5
13	Low	PWM1	SD5	DPI_D9	SPI5_MISO	RXD5	SCL5
14	Low	TXD0	SD6	DPI_D10	SPI5_MOSI	CTS5	TXD1
15	Low	RXD0	SD7	DPI_D11	SPI5_SCLK	RTS5	RXD1
16	Low	FL0	SD8	DPI_D12	CTS0	SPI1_CE2_N	CTS1
17	Low	FL1	SD9	DPI_D13	RTS0	SPI1_CE1_N	RTS1
18	Low	PCM_CLK	SD10	DPI_D14	SPI6_CE0_N	SPI1_CE0_N	PWM0
19	Low	PCM_FS	SD11	DPI_D15	SPI6_MISO	SPI1_MISO	PWM1
20	Low	PCM_DIN	SD12	DPI_D16	SPI6_MOSI	SPI1_MOSI	GPCLK0
21	Low	PCM_DOUT	SD13	DPI_D17	SPI6_SCLK	SPI1_SCLK	GPCLK1
22	Low	SD0_CLK	SD14	DPI_D18	SD1_CLK	ARM_TRST	SDA6
23	Low	SD0_CMD	SD15	DPI_D19	SD1_CMD	ARM_RTCK	SCL6
24	Low	SD0_DAT0	SD16	DPI_D20	SD1_DAT0	ARM_TDO	SPI3_CE1_N
25	Low	SD0_DAT1	SD17	DPI_D21	SD1_DAT1	ARM_TCK	SPI4_CE1_N
26	Low	SD0_DAT2	TE0	DPI_D22	SD1_DAT2	ARM_TDI	SPI5_CE1_N
27	Low	SD0_DAT3	TE1	DPI_D23	SD1_DAT3	ARM_TMS	SPI6_CE1_N

Table 5: Raspberry Pi 4 GPIO Alternate Functions

Table 5 details the default pin pull state and available alternate GPIO functions. Most of these alternate peripheral functions are described in detail in the BCM2711 Peripherals Specification document which can be downloaded from the [hardware documentation](#) section of the website.



5.1.3 Display Parallel Interface (DPI)

A standard parallel RGB (DPI) interface is available the GPIOs. This up-to-24-bit parallel interface can support a secondary display.

5.1.4 SD/SDIO Interface

The Pi4B has a dedicated SD card socket which supports 1.8V, DDR50 mode (at a peak bandwidth of 50 Megabytes / sec). In addition, a legacy SDIO interface is available on the GPIO pins.

5.2 Camera and Display Interfaces

The Pi4B has 1x Raspberry Pi 2-lane MIPI CSI Camera and 1x Raspberry Pi 2-lane MIPI DSI Display connector. These connectors are backwards compatible with legacy Raspberry Pi boards, and support all of the available Raspberry Pi camera and display peripherals.

5.3 USB

The Pi4B has 2x USB2 and 2x USB3 type-A sockets. Downstream USB current is limited to approximately 1.1A in aggregate over the four sockets.

5.4 HDMI

The Pi4B has 2x micro-HDMI ports, both of which support CEC and HDMI 2.0 with resolutions up to 4Kp60.

5.5 Audio and Composite (TV Out)

The Pi4B supports near-CD-quality analogue audio output and composite TV-output via a 4-ring TRS 'A/V' jack.

The analog audio output can drive 32 Ohm headphones directly.

5.6 Temperature Range and Thermals

The recommended ambient operating temperature range is 0 to 50 degrees Celsius.

To reduce thermal output when idling or under light load, the Pi4B reduces the CPU clock speed and voltage. During heavier load the speed and voltage (and hence thermal output) are increased. The internal governor will throttle back both the CPU speed and voltage to make sure the CPU temperature never exceeds 85 degrees C.

The Pi4B will operate perfectly well without any extra cooling and is designed for sprint performance - expecting a light use case on average and ramping up the CPU speed when needed (e.g. when loading a webpage). If a user wishes to load the system continually or operate it at a high temperature at full performance, further cooling may be needed.

6 Availability

Raspberry Pi guarantees availability of the Pi4B until at least January 2031.

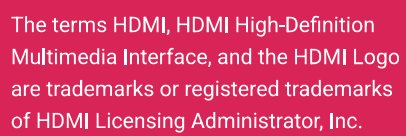


7 Support

For support please see the hardware documentation section of the [Raspberry Pi website](#) and post questions to the [Raspberry Pi forum](#).



A detailed 3D line-art illustration of a computer motherboard, viewed from an isometric perspective. The motherboard is populated with various components: a large central CPU, several rectangular RAM modules, and multiple expansion slots including PCI and SATA. A long row of pins is visible along one edge, likely for a system bus or connector. The entire assembly is rendered in a clean, minimalist style with black outlines on a white background.



Overview



Welcome to the latest generation of Raspberry Pi: the everything computer.

Featuring a 64-bit quad-core Arm Cortex-A76 processor running at 2.4GHz, Raspberry Pi 5 delivers a 2–3× increase in CPU performance relative to Raspberry Pi 4. Alongside a substantial uplift in graphics performance from an 800MHz VideoCore VII GPU; dual 4Kp60 display output over HDMI; and state-of-the-art camera support from a rearchitected Raspberry Pi Image Signal Processor, it provides a smooth desktop experience for consumers, and opens the door to new applications for industrial customers.

For the first time, this is a full-size Raspberry Pi computer using silicon built in-house at Raspberry Pi. The RP1 “southbridge” provides the bulk of the I/O capabilities for Raspberry Pi 5, and delivers a step change in peripheral performance and functionality. Aggregate USB bandwidth is more than doubled, yielding faster transfer speeds to external UAS drives and other high-speed peripherals; the dedicated two-lane 1Gbps MIPI camera and display interfaces present on earlier models have been replaced by a pair of four-lane 1.5Gbps MIPI transceivers, tripling total bandwidth, and supporting any combination of up to two cameras or displays; peak SD card performance is doubled, through support for the SDR104 high-speed mode; and for the first time the platform exposes a single-lane PCI Express 2.0 interface, providing support for high-bandwidth peripherals.

Specification

Processor: Broadcom BCM2712 2.4GHz quad-core 64-bit Arm Cortex-A76 CPU, with Cryptographic Extension, 512KB per-core L2 caches, and a 2MB shared L3 cache

Features:

- VideoCore VII GPU, supporting OpenGL ES 3.1, Vulkan 1.2
- Dual 4Kp60 HDMI® display output with HDR support
- 4Kp60 HEVC decoder
- LPDDR4X-4267 SDRAM (options for 1GB, 2GB, 4GB, 8GB and 16GB)
- Dual-band 802.11ac Wi-Fi®
- Bluetooth 5.0/Bluetooth Low Energy (BLE)
- microSD card slot, with support for high-speed SDR104 mode
- 2 × USB 3.0 ports, supporting simultaneous 5Gbps operation
- 2 × USB 2.0 ports
- Gigabit Ethernet, with PoE+ support (requires separate PoE+ HAT)
- 2 × 4-lane MIPI camera/display transceivers
- PCIe 2.0 x1 interface for fast peripherals (requires separate M.2 HAT or other adapter)
- 5V/5A DC power via USB-C, with Power Delivery support
- Raspberry Pi standard 40-pin header
- Real-time clock (RTC), powered from external battery
- Power button

MTBF¹ Ground Benign: 93 800 hours

Operating temperature: 0°C to 70°C

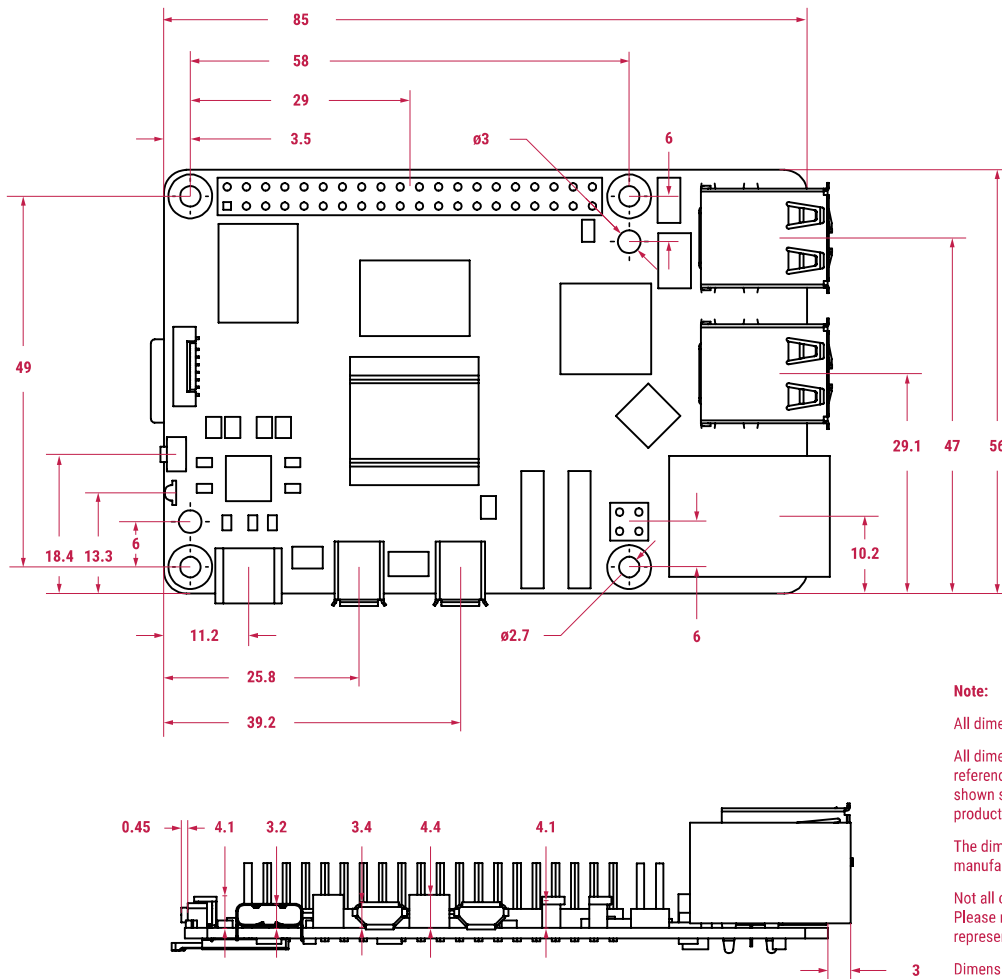
Production lifetime: Raspberry Pi 5 will remain in production until at least January 2036

Compliance: For a full list of local and regional product approvals, please visit pip.raspberrypi.com

List price:	1GB	\$45
	2GB	\$65
	4GB	\$110
	8GB	\$175
	16GB	\$305

¹ Mean Time Between Failure

Physical specification



Note:

All dimensions in mm

All dimensions are approximate and for reference purposes only. The dimensions shown should not be used for producing production data

The dimensions are subject to part and manufacturing tolerances

Not all of the board components are shown. Please reference a physical board for representation of componentry

Dimensions may be subject to change

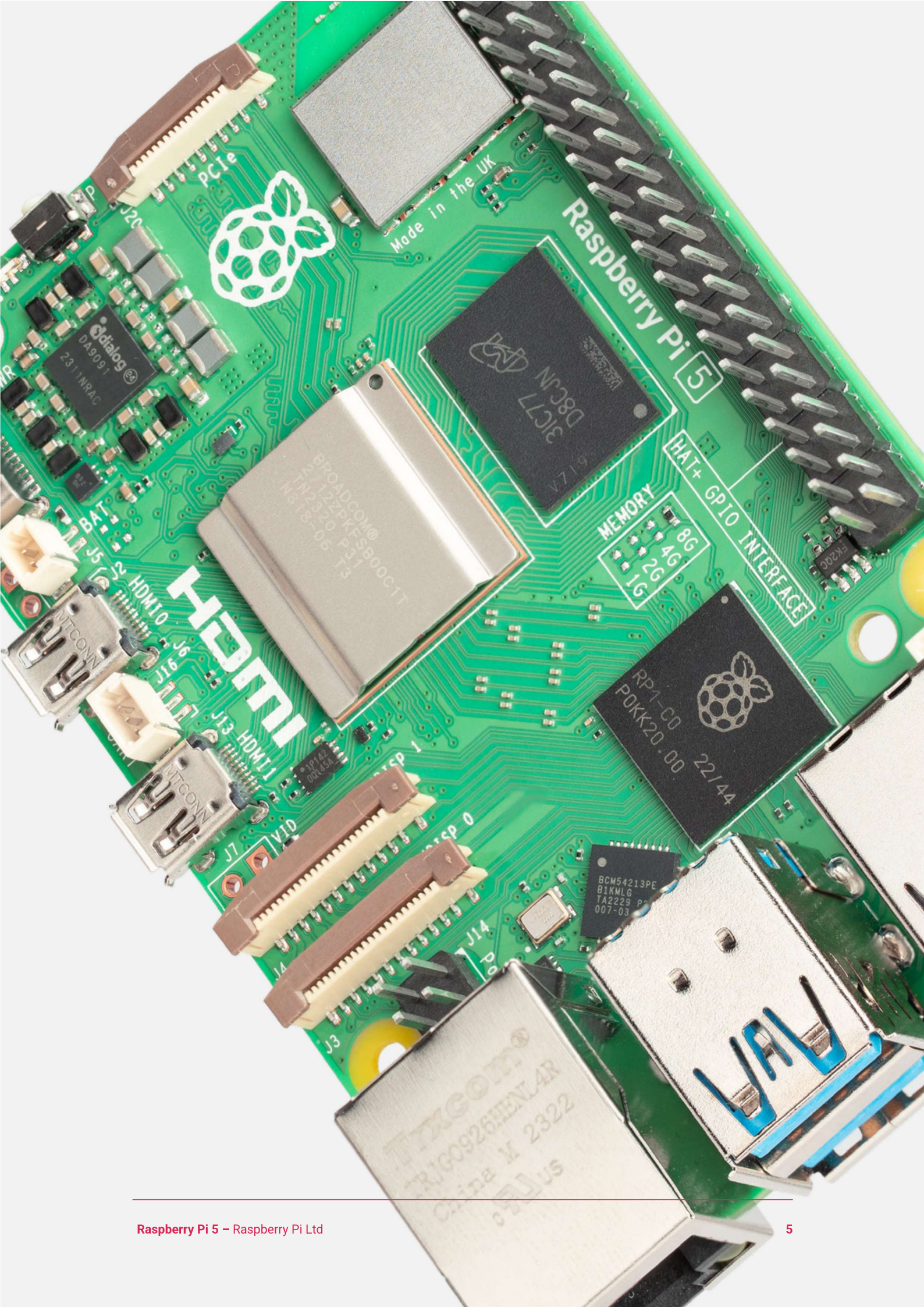
WARNINGS

- This product should be operated in a well ventilated environment, and if used inside a case, the case should not be covered.
- While in use, this product should be firmly secured or should be placed on a stable, flat, non-conductive surface, and should not be contacted by conductive items.
- The connection of incompatible devices to Raspberry Pi 5 may affect compliance, result in damage to the unit, and invalidate the warranty.
- All peripherals used with this product should comply with relevant standards for the country of use and be marked accordingly to ensure that safety and performance requirements are met.

SAFETY INSTRUCTIONS

To avoid malfunction or damage to this product, please observe the following:

- Do not expose to water or moisture, or place on a conductive surface while in operation.
- Do not expose to heat from any source; Raspberry Pi 5 is designed for reliable operation at normal ambient temperatures.
- Store in a cool, dry location.
- Take care while handling to avoid mechanical or electrical damage to the printed circuit board and connectors.
- While it is powered, avoid handling the printed circuit board, or handle it only by the edges, to minimise the risk of electrostatic discharge damage.



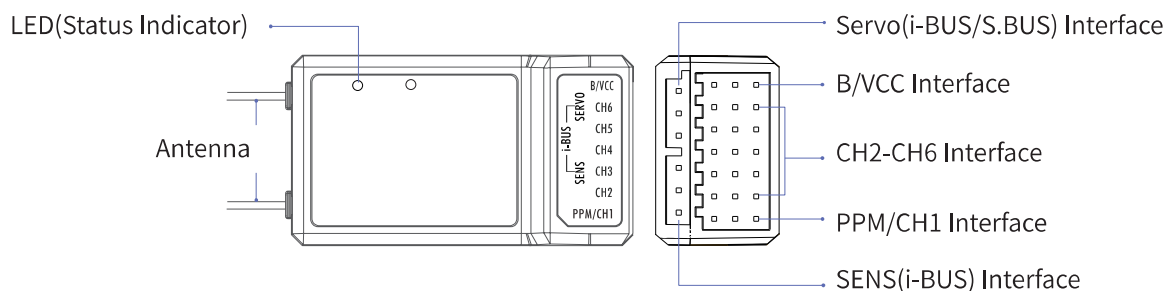


Raspberry Pi is a trademark of Raspberry Pi Ltd

Introduction

FS-iA6B is a two-way receiver for multicopters, helicopters, and fixed-wing aircrafts, adopting AFHDS 2A (second-generation enhanced automatic frequency hopping digital system). It has 8 channels and can output standard PPM signal, PWM, i-BUS or S.BUS signal.

Receiver Overview



Interface introduction

PPM/CH1 Interface: Connects servo or output PPM signal.

CH 2 ~ CH6: Connects to servos, power supply or other components.

B/VCC: Used to connect the binding cable during binding, and used to connect the power cord during normal operation.

Servo(i-BUS/S.BUS) Interface: Used to connect i-BUS receiver to expand channels. It can output i-BUS or S.BUS signal.

SENS(i-BUS) Interface: For connecting i-BUS sensors.

Product Specifications

- Product Model: FS-iA6B
- Number of PWM Channels: 6
- Compatible Models: Multicopters, Fixed-wing Aircrafts, Helicopters
- Compatible Transmitters: FS-i6S
- Data Output: PWM/PPM/i-BUS/S.BUS
- Frequency Range: 2.408-2.475GHz
- Bandwidth: 500KHz
- Number of Bands: 135
- Maximum Power: < 20dBm (e.i.r.p.) (EU)
- Receiver Sensitivity: - 92dBm
- RF Protocol : AFHDS 2A
- Modulation Type: GFSK
- Transmission Method: FHSS
- Channel Latency: <15ms
- Antenna: Two Antennas
- Input Power: 4.0-8.4V/DC
- Display: LED Indicator
- Online Update: None
- Distance : More than 300m(Ground distance without interference)
- Dimensions: 47 * 26.2 * 15 mm
- Weight: 14.9 g
- Certification: CE, RCM, FCC ID:N4ZFLYSKYIA6B

Binding

The receiver supports two -way binding. The transmitter will display the information returned from the receiver after the binding is completed.

Follow the steps below to bind.

1. Put the transmitter into binding state.
2. Plug the binding cable into the B/VCC interface of the receiver.
3. Plug the power cord into any other interface. At the same time, the receiver LED flashes quickly, indicating that it has entered the binding state.
4. The receiver LED is solid on, indicating the binding is successful.
5. Unplug the binding cable from the receiver.
6. Check to make sure the transmitter and receiver functions are working correctly, repeat steps 1 to 5 (binding process) if any problems arise.

! Attention:

- Make sure the product is installed and calibrated correctly, failure to do so may result in serious injury.
- Make sure the receiver's battery is disconnected before turning off the transmitter, failure to do so can result out of control. Unreasonable setting of the Failsafe may cause accidents.
- Make sure the receiver is mounted away from motors, electronic speed controllers or any device that emits excessive electrical noise.
- Keep the receiver's antenna at least 1cm away from conductive materials such as carbon or metal.
- Do not power on the receiver during the setup process to prevent loss of control.

Certifications

FCC Compliance Statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Warning: changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

EU DoC Declaration

Hereby, [ShenZhen FLYSKY Technology Co., Ltd.] declares that the Radio Equipment [FS-iA6B] is in compliance with RED 2014/53/EU. The full text of the EU DoC is available at the following internet address: www.flyskytech.com/info_detail/10.html

RF Exposure Compliance

The distance between user and products should be no less than 20cm.

Environmentally friendly disposal

Old electrical appliances must not be disposed of together with the residual waste, but have to be disposed of separately. The disposal at the communal collecting point via private persons is for free. The owner of old appliances is responsible to bring the appliances to these collecting points or to similar collection points. With this little personal effort, you contribute to recycle valuable raw materials and the treatment of toxic substances.



FCC ID: N4ZFLYSKYIA6B



微信公众号



Bilibili



Website



Facebook

Manufacturer: ShenZhen FLYSKY Technology Co., Ltd.

Address: 16F, Huafeng Building, No. 6006 Shennan Road, Futian District, Shenzhen, Guangdong, China

Figures and illustrations in this manual are provided for reference only and may differ from actual product appearance. Product design and specifications may be changed without notice.

<http://www.flysky-cn.com>

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SpeedyBee

F405 V4 BLS 55A 30x30 Stack

User Manual V1.0

Contents (Click on any section to jump)

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Part 2 – SpeedyBee F405 V4 Flight Controller

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Part 3 – SpeedyBee BLS 55A 4-in-1 ESC

▪ Layout	10
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▪ ESC Configuration	12
▪ ESC Firmware Update	13
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Part 1 - Overview

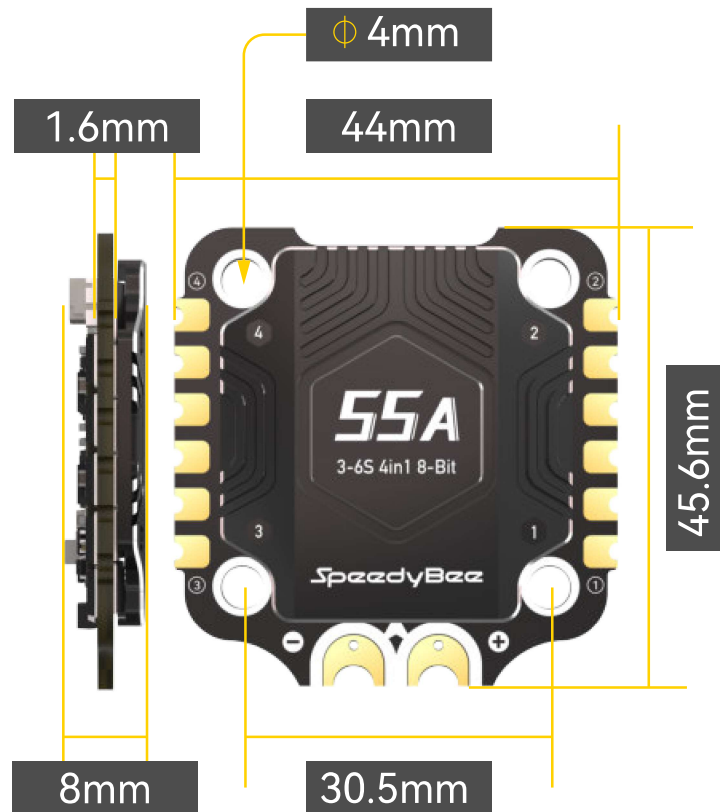
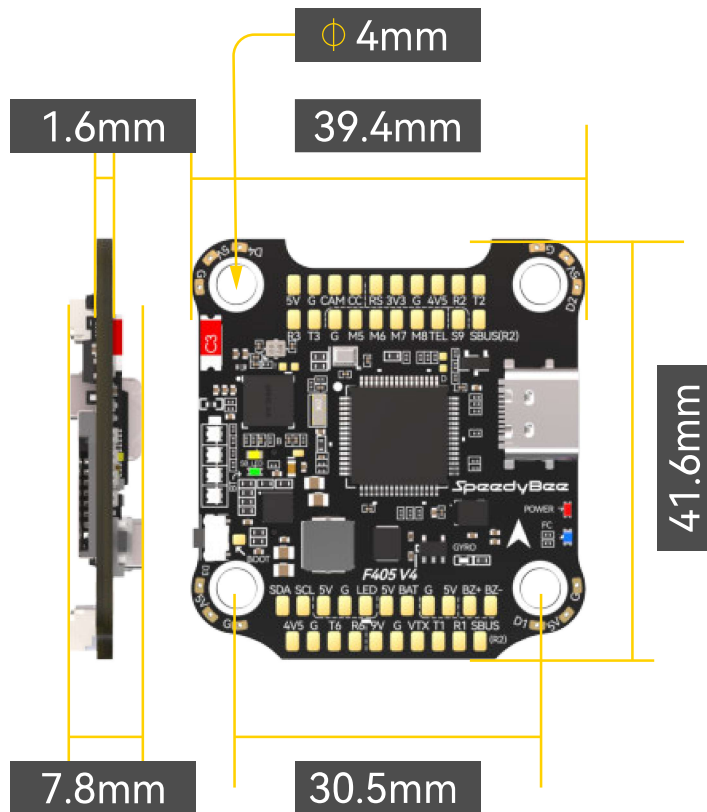
Specs Overview

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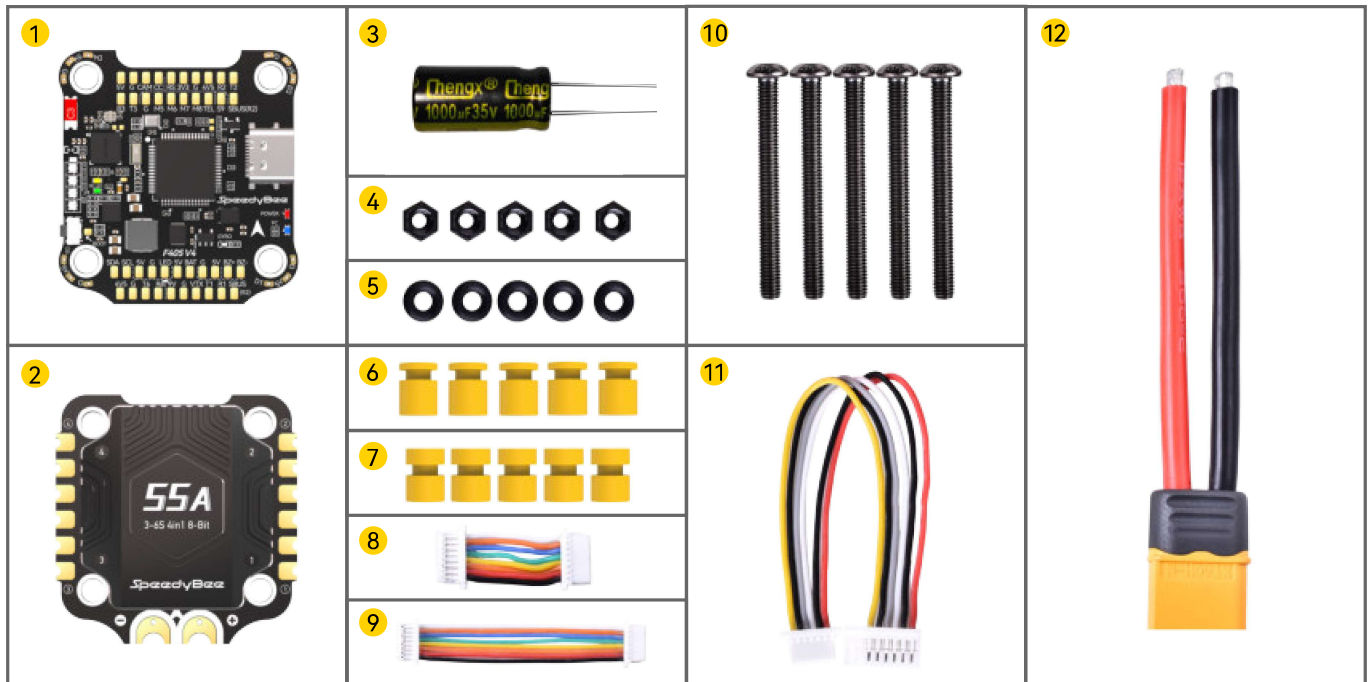
Product Name	SpeedyBee F405 V4 BLS 55A 30x30 Stack
Flight Controller	SpeedyBee F405 V4 Flight Controller
ESC	SpeedyBee BLS 55A 4-in-1 ESC
Bluetooth	Supported. For FC & ESC parameter settings
Wireless FC Firmware Flashing	NOT Supported
Wireless Blackbox Dwonload & Analysis	NOT Supported
Power Input	3-6S LiPo
Mounting	30.5 x 30.5mm (4mm hole size)
Dimension	45.6mm(L) x 44mm(W) x 18.3mm(H)
Weight	34g

Dimensions

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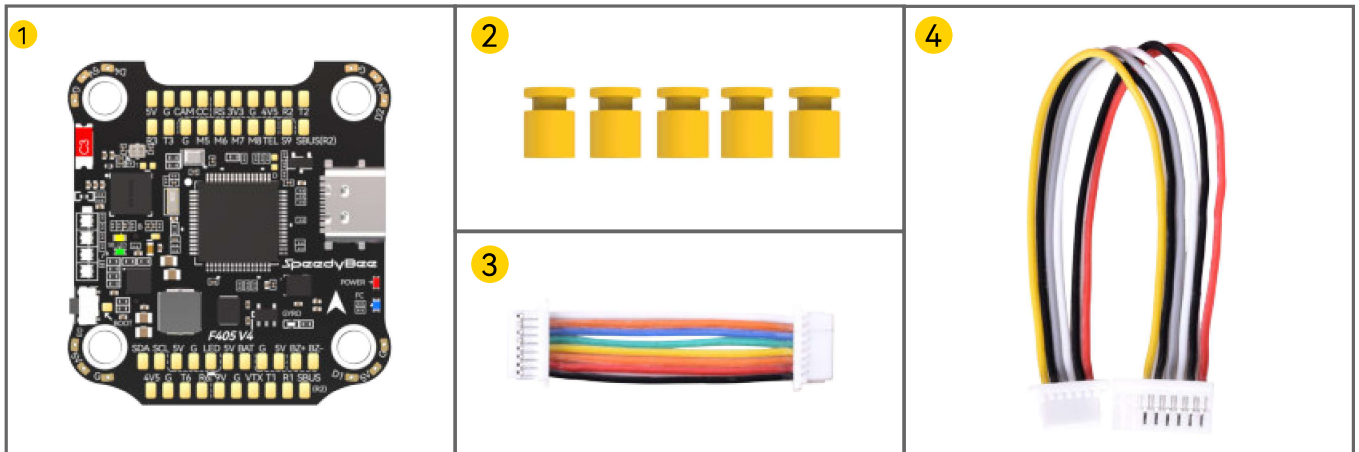
Option 1 – SpeedyBee F405 V4 55A 30x30 Stack



- 1 SpeedyBee F405 V4 Flight Controller x 1
- 2 SpeedyBee BLS 55A 4-in-1 ESC x 1
- 3 35V 1000uF Low ESR Capacitor x 1
- 4 M3 Nylon Nut x 5
- 5 M3 silicone O Ring x 5
- 6 M3*8mm Silicone Grommets(for FC) x 5
- 7 M3*8.1mm Silicone Grommets(for ESC) x 5
- 8 SH 1.0mm 15mm-length 8pin Cable(for FC-ESC connection) x 1
- 9 SH 1.0mm 75mm-length 8pin Cable* x 1
- 10 M3*30mm Iner-hexagon Screws x 5
- 11 DJI 6pin Cable(80mm) x 1
- 12 XT60 Power Cable(70mm) x 1

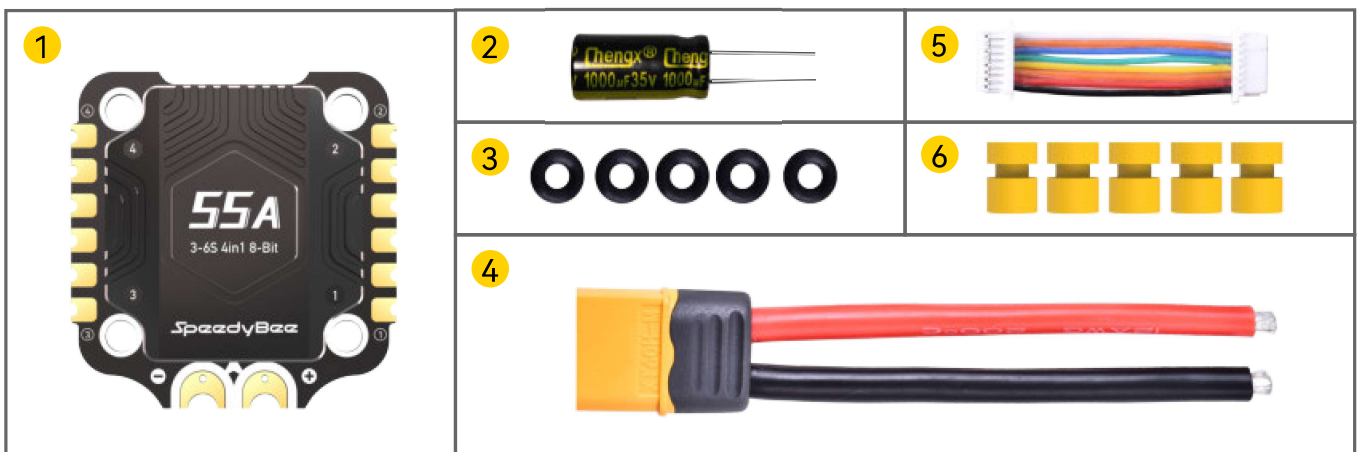
* When the ESC tail faces the drone's front, use this cable for the flight controller and ESC; both ends plug interchangeably.

Option 2 – SpeedyBee F405 V4 Flight Controller



- 1 SpeedyBee F405 V4 Flight Controller x 1
- 2 M3*8mm Silicone Grommets(for FC) x 5
- 3 SH 1.0mm 30mm-length 8pin Cable(for FC-ESC connection) x 1
- 4 DJI 6pin Cable(80mm) x 1

Option 3 – SpeedyBee BLS 55A 4-in-1 ESC

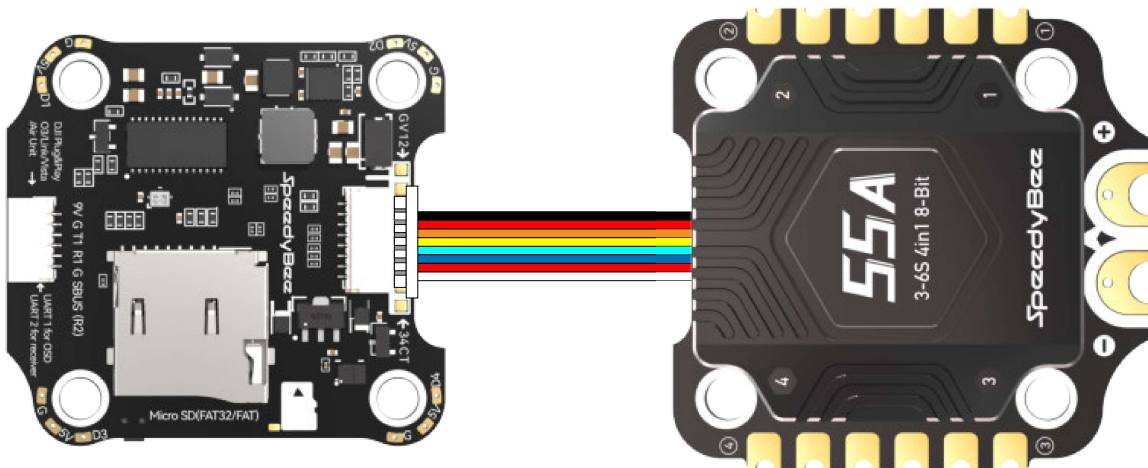


- 1 SpeedyBee BLS 55A 4-in-1 ESC x 1
- 2 35V 1000uF Low ESR Capacitor x 1
- 3 M3 silicone O Ring x 5
- 4 XT60 Power Cable(70mm) x 1
- 5 SH 1.0mm 30mm-length 8pin Cable(for FC-ESC connection) x 1
- 6 M3*8.1mm Silicone Grommets(for ESC) x 5

Use the 8-pin cable in the package to connect the FC and the ESC.
Or solder 8 wires directly to the 8 pads on each end.

Method 1 - Using 8-pin cable

Use any end of the 8-pin JST cable to connect the FC to the ESC.

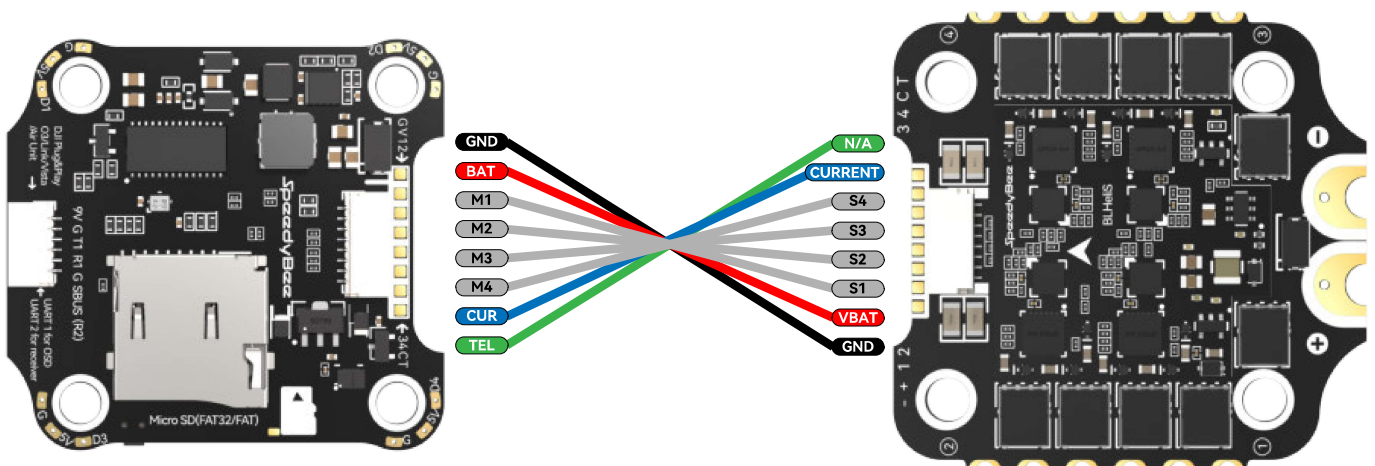


F405 V4 Flight Controller

BLS 55A 4-in-1 ESC

Method 2 - Direct soldering

Solder 8 wires to the 8 pads on each end referring to the pad definition below.



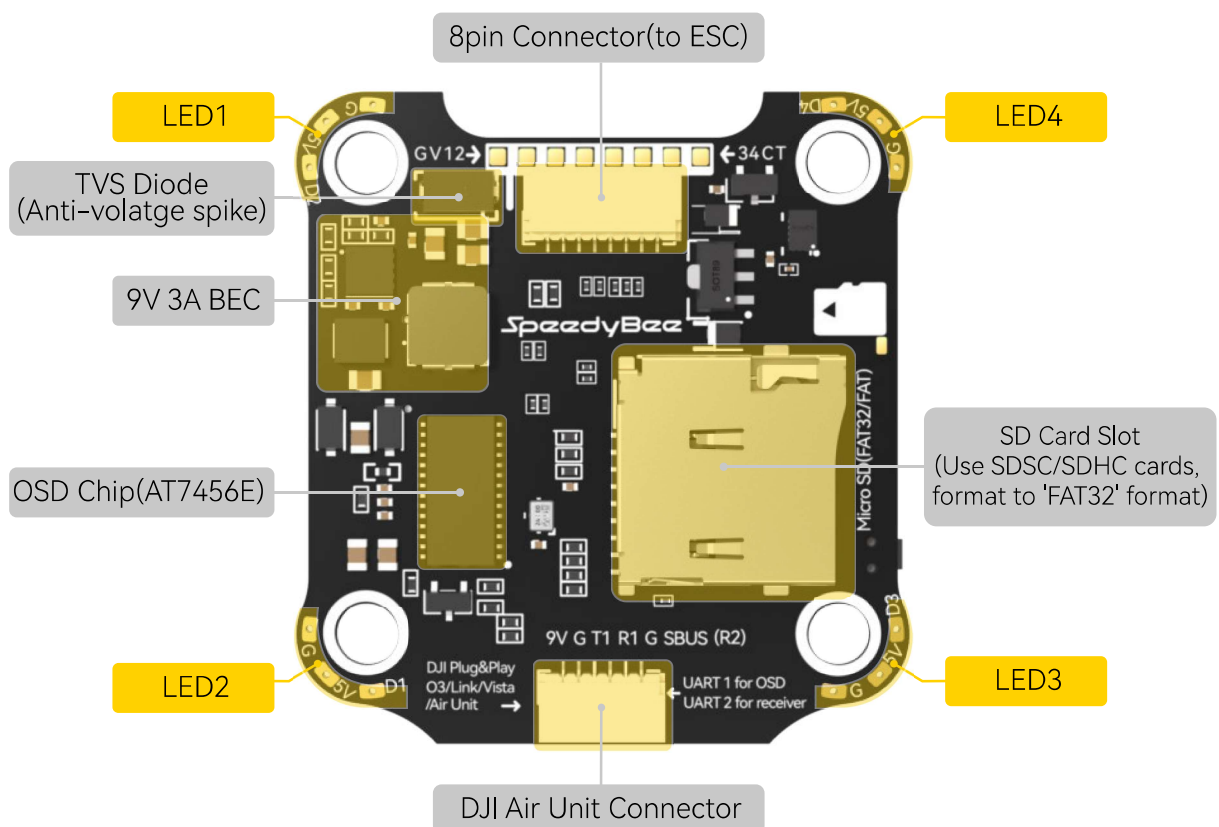
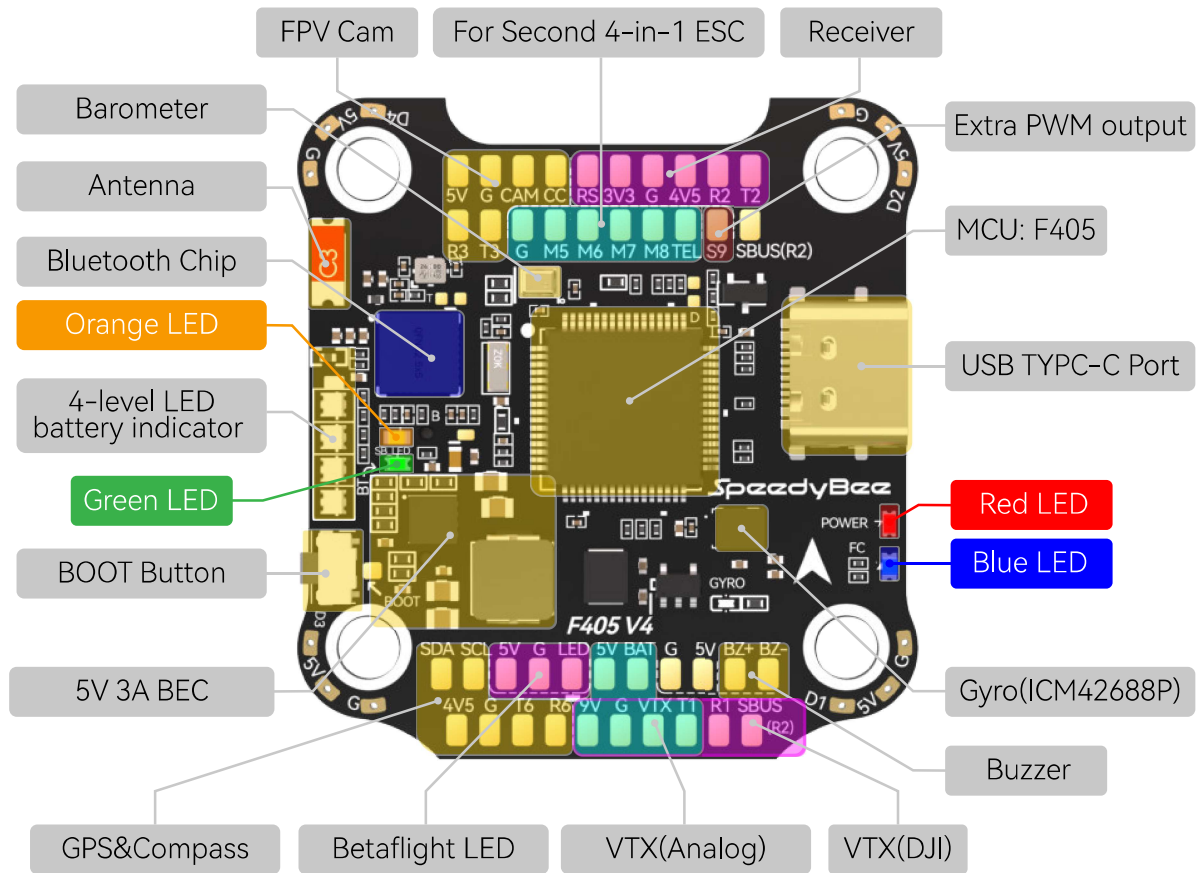
F405 V4 Flight Controller

BLS 55A 4-in-1 ESC

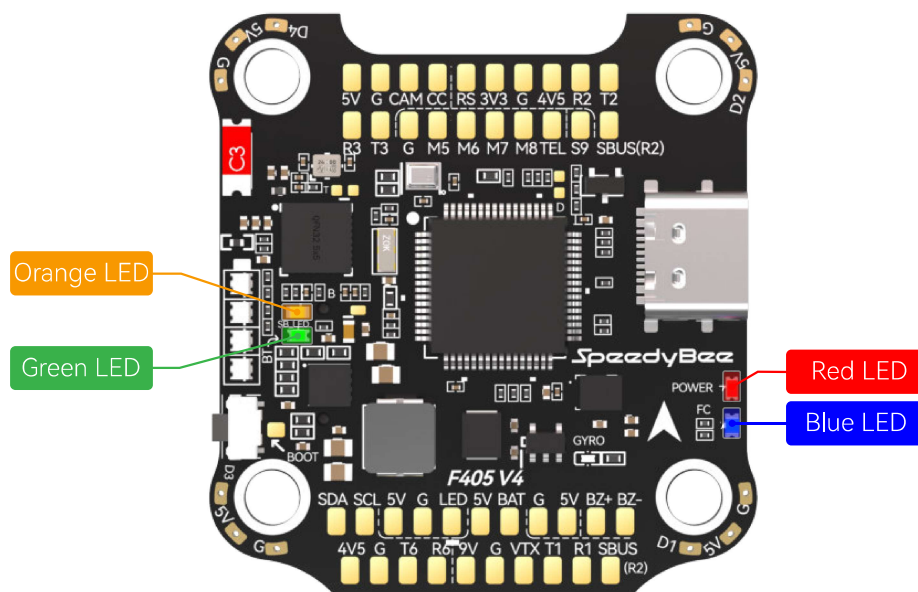
Part 2 - F405 V4 Flight Controller

Layout

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■ LED Indicator Definition



■ **RED LED** - Power Indicator. **Solid Red** after powering up.

■ **GREEN LED** - Bluetooth status light. **Solid Green** indicates Bluetooth is connected.

■ **BLUE LED** - Flight controller status light which is controlled by the flight controller firmware.

■ **Orange LED** - LED Control Mode Indicator. It indicates the 4 sets of LED strips connected to LED1-LED4 pads on the corners of the flight controller are controlled by Betaflight firmware(BF_LED mode) or the Bluetooth chip(SB_LED mode).

Solid Orange : indicates the 4 x LEDs are in SB_LED mode. In this mode, when the FC is powered on and in standby mode, press the BOOT button to cycle the display modes of the LEDs.

OFF : indicates the 4 x LEDs are controlled by Betaflight firmware.

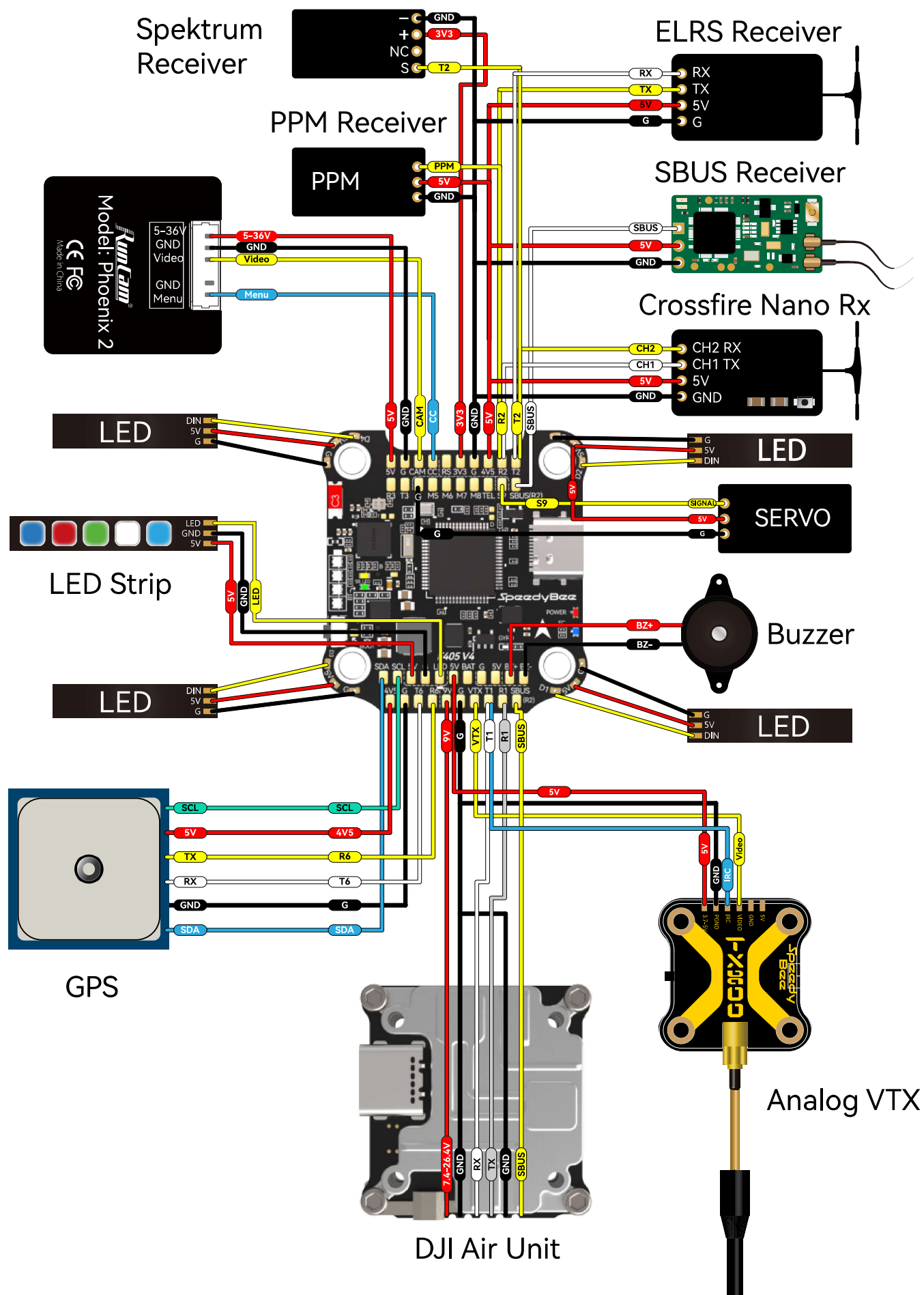
Long press the button for 3 seconds to switch the control modes between BF_LED mode and SB_LED mode.

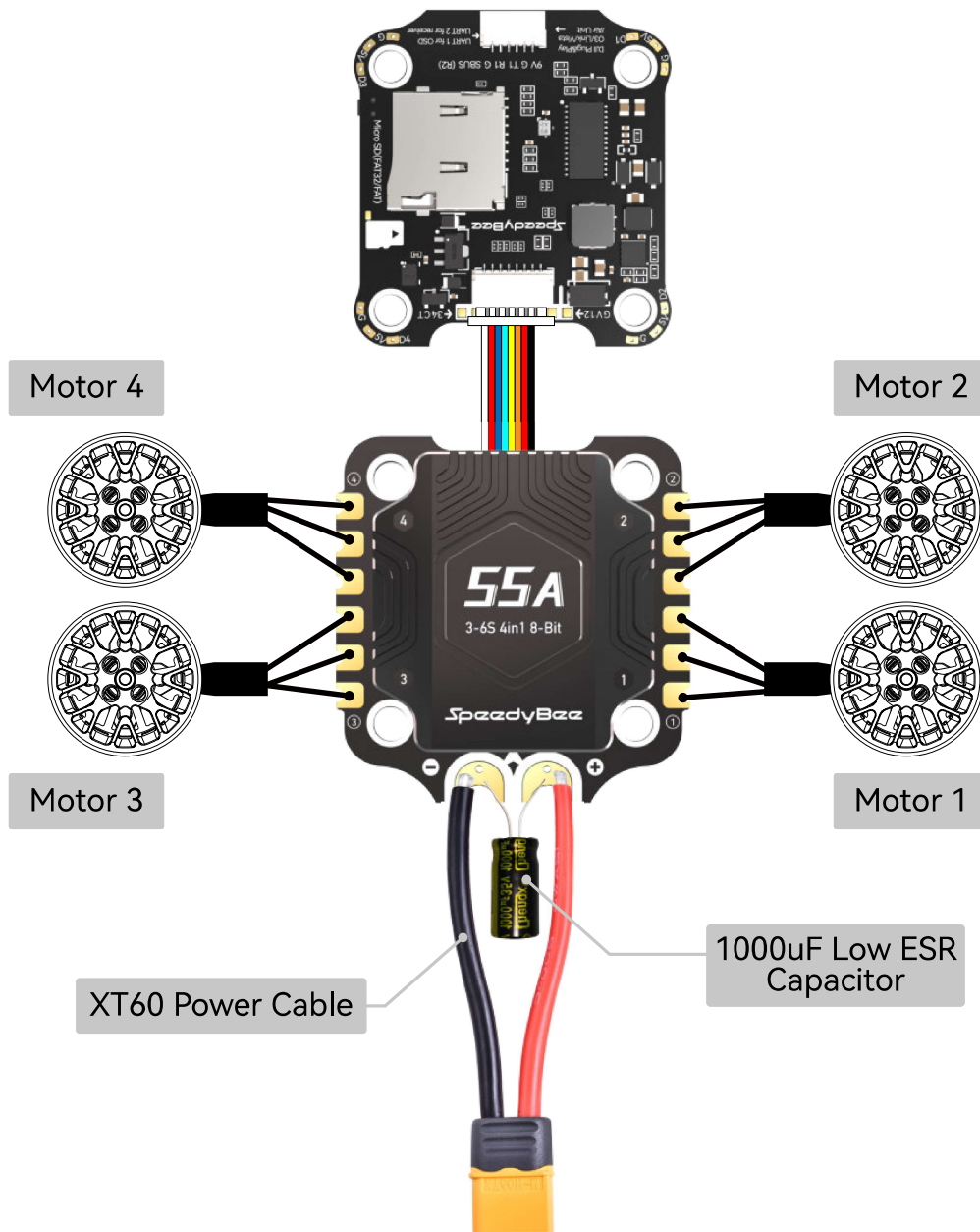
■ BOOT Button

[A]Only if the flight controller gets bricked and can't power up, please follow these steps to re-flash firmware for it:

- ① Insert a USB A to TYPE-C cable to your PC.
- ② Press and hold the BOOT button, insert the USB cable into the flight controller, then release the BOOT button.
- ③ Open Betaflight/INAV configurator on the PC, go to the 'Firmware Flashing' page, choose the target 'SPEEDYBEEF405V4' and flash.

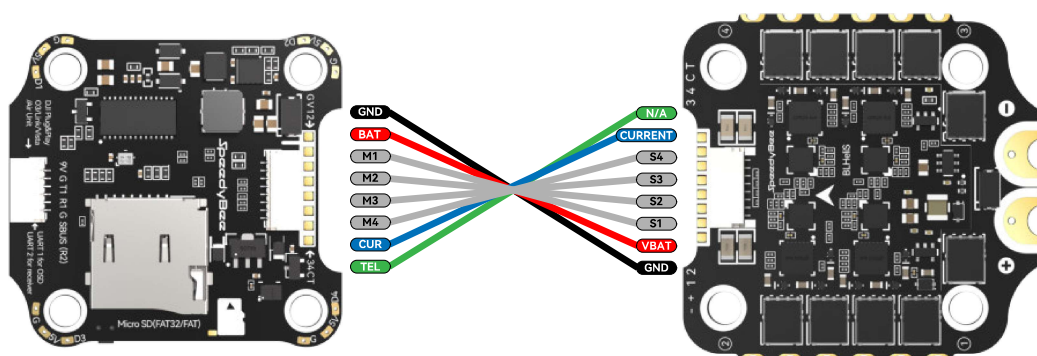
[B]. When the FC is powered on and in standby mode, the BOOT button can be used to control the LED strips connected to LED1-LED4 pads on the corners. Short-press the BOOT button to cycle the LED displaying mode. Long-press the BOOT button to switch between SpeedyBee-LED mode and BF-LED mode. Under BF-LED mode, all the LED1-LED4 strips will be controlled by the Betaflight firmware. The default mode is BF-LED mode.





Note1: In order to prevent the stack from being burnt out by voltage spikes on powering up, it is strongly recommended to use the Low ESR capacitor in the package.

Note2: The FC and ESC can also connected via direct soldering. Soldering pads definition is as follows.

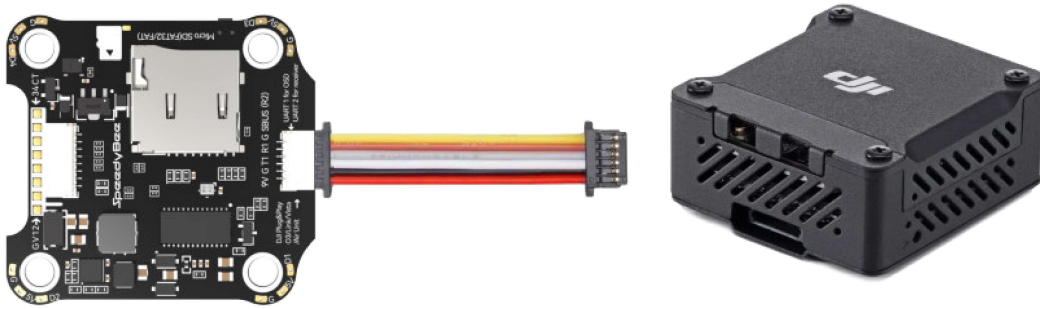


F405 V4 Flight Controller

BLS 55A 4-in-1 ESC

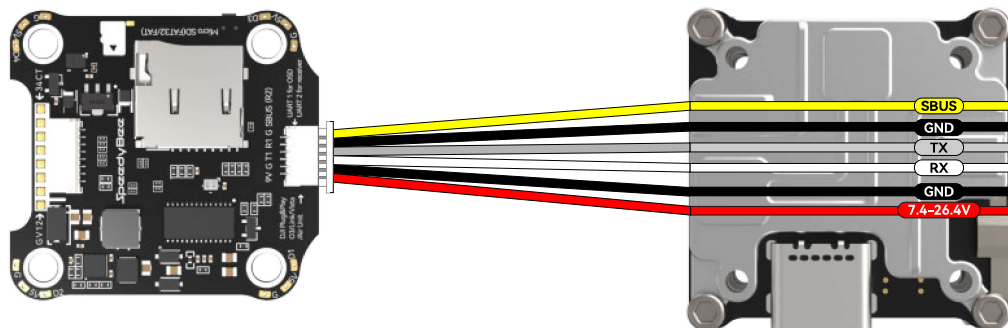
Cable Connection vs DJI O3 Air Unit

Use 6-pin cable comes with the O3 Air Unit



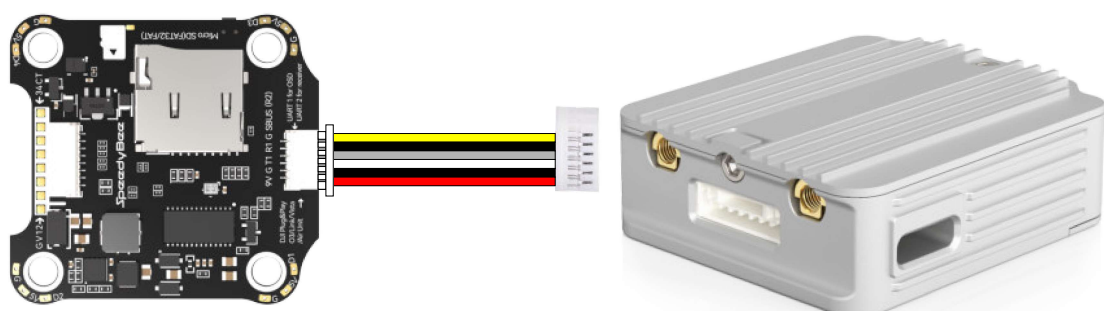
Cable Connection vs RunCam Link/Caddx Vista Air Unit

Use 6-pin cable comes with the F405 V4 stack
(See the accessory No.11 in the package section)



Cable Connection vs DJI Air Unit V1

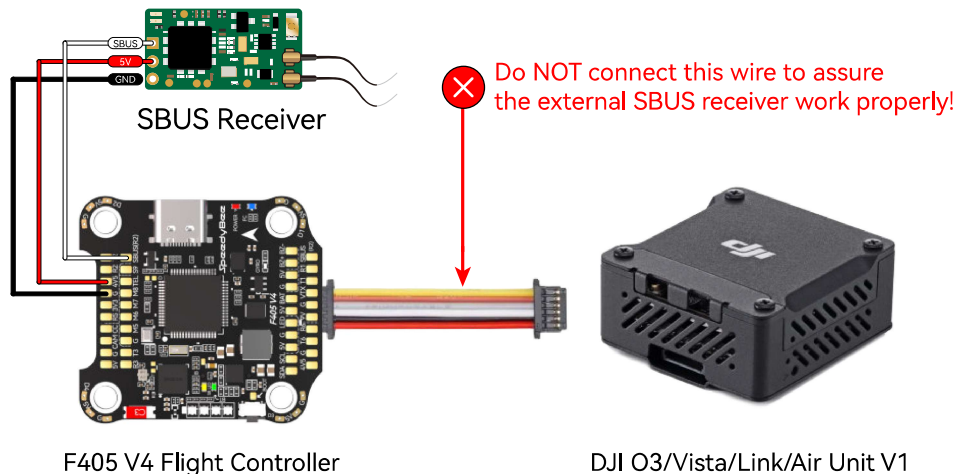
Use 6-pin cable comes with the F405 V4 stack
(See the accessory No.11 in the package section)



Importance notice for SBUS receiver

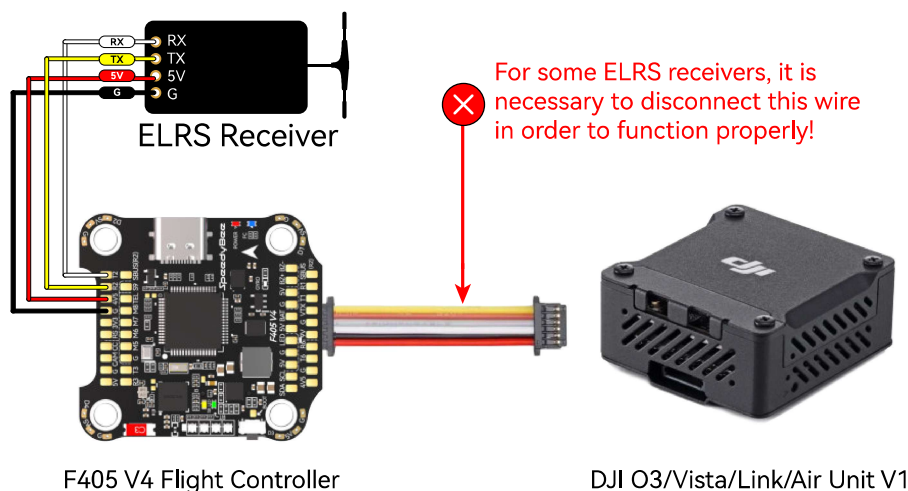
When using an SBUS receiver, the SBUS signal wire of the receiver must be connected to the SBUS pad on the front side of the flight controller (this pad internally uses UART2).

If you are also using the DJI Air Unit(O3/Link/Vista/Air Unit V1), you will need to disconnect the SBUS signal wire from the Air Unit harness. Failure to do so will prevent the SBUS receiver from being properly recognized by the flight controller. You can use tweezers to pick out the SBUS wire from the 6-pin harness connector (or directly cut this wire) and insulate the exposed part of the wire carefully.



Importance notice for ELRS receiver

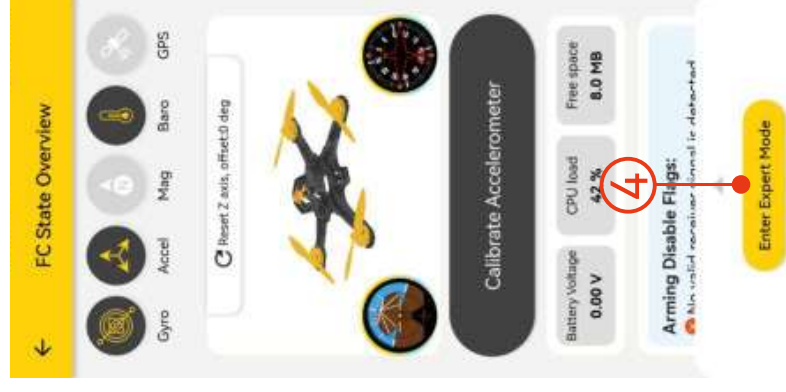
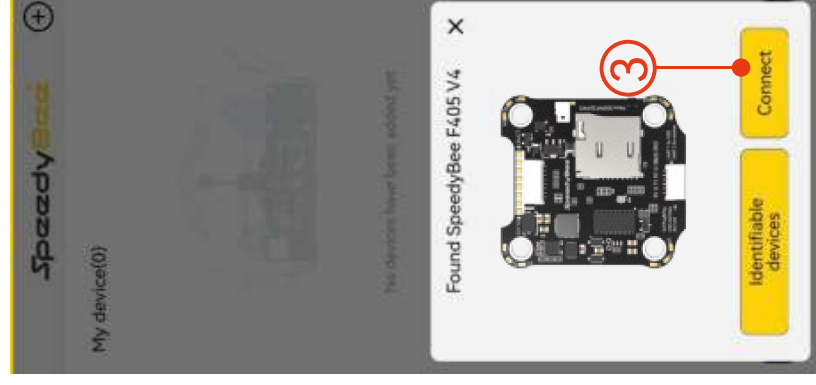
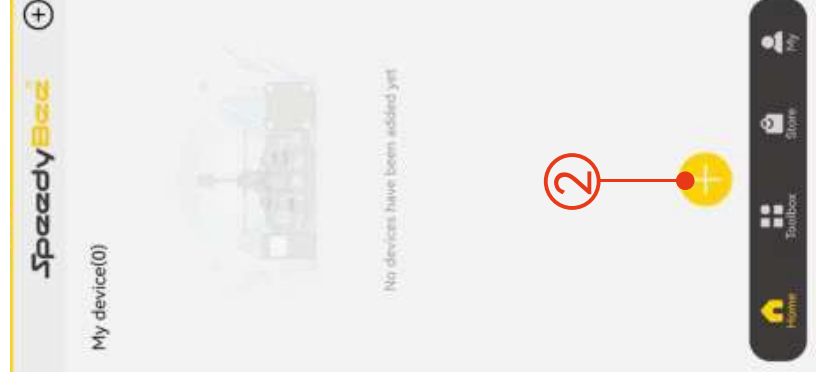
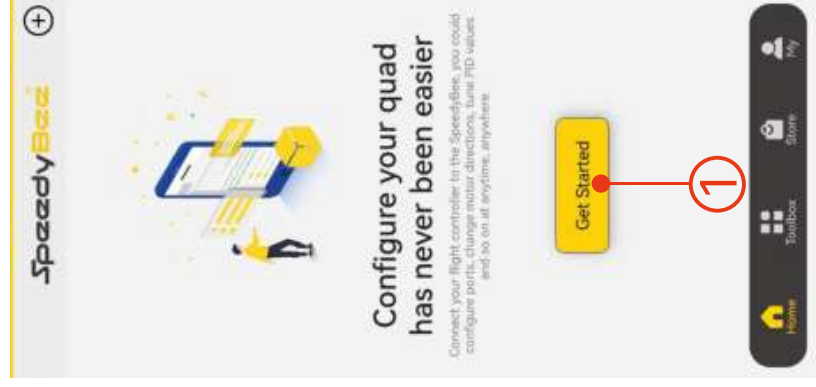
We recommend connecting the ELRS receiver's TX and RX to the T2 and R2 pads on the flight controller. However, when using the DJI Air Unit simultaneously, some ELRS receivers may not be recognized properly by the flight controller. If you encounter this issue, you need to disconnect the SBUS signal wire from the Air Unit harness. You can use tweezers to pick out the SBUS wire from the 6-pin harness connector (or directly cut this wire) and insulate the exposed part of the wire carefully.



■ Get the SpeedyBee App

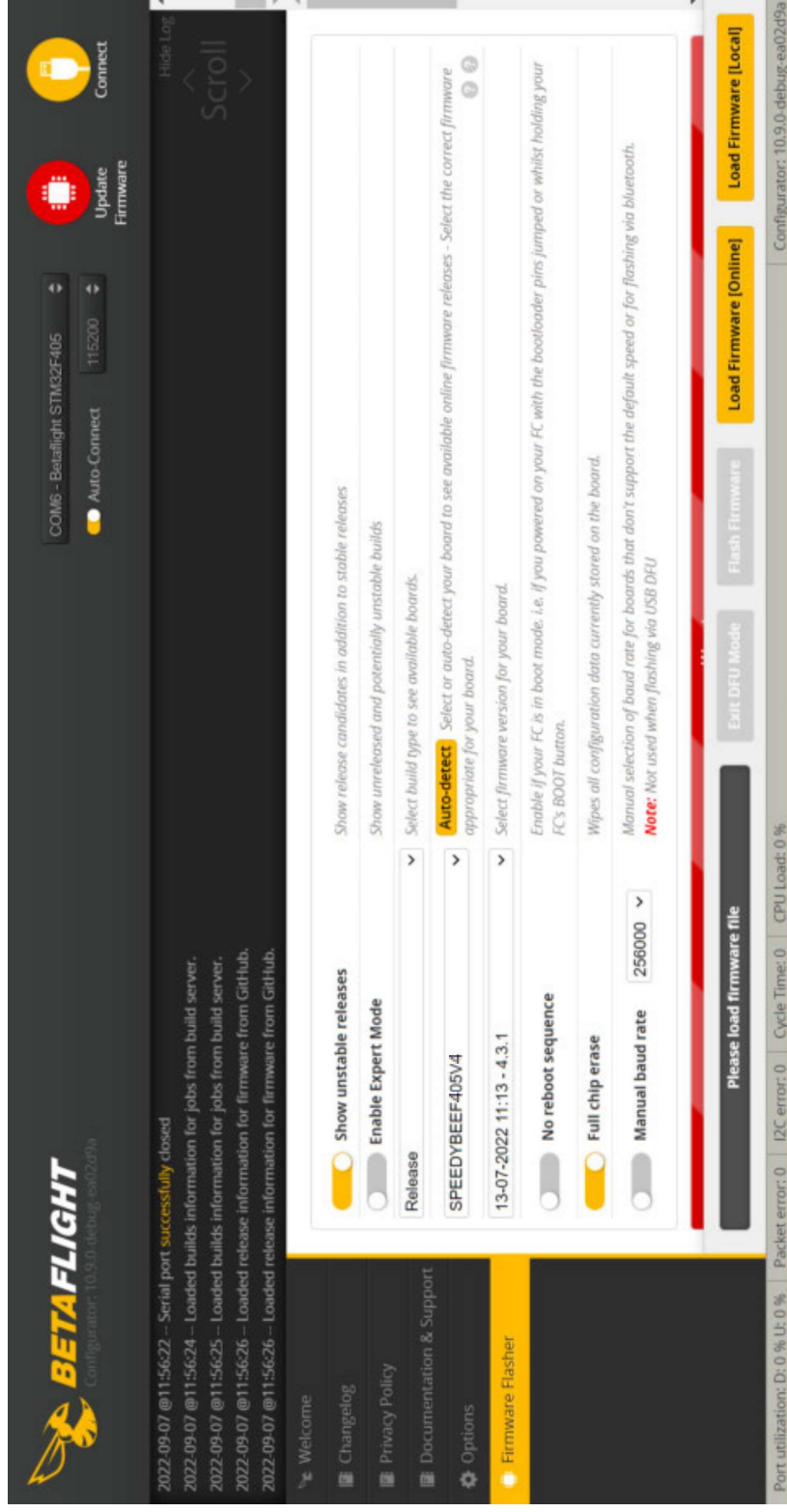
Search 'SpeedyBee' on Google Play or App Store. Or download the Android .apk file on our website: <https://www.speedybee.com/download>.

■ FC Configuration



SpeedyBee F405 V4 flight controller does not support wireless firmware flashing, so please flash firmware for it on your PC following the steps below:

- ① Connect the flight controller to the PC with a USB cable
- ② Open Betaflight/ INAV configurator on your PC. Take Betaflight configurator as an example, go to the 'Firmware Flashing' page, choose the target '**SPEEDYBEEF405V4**' and flash.

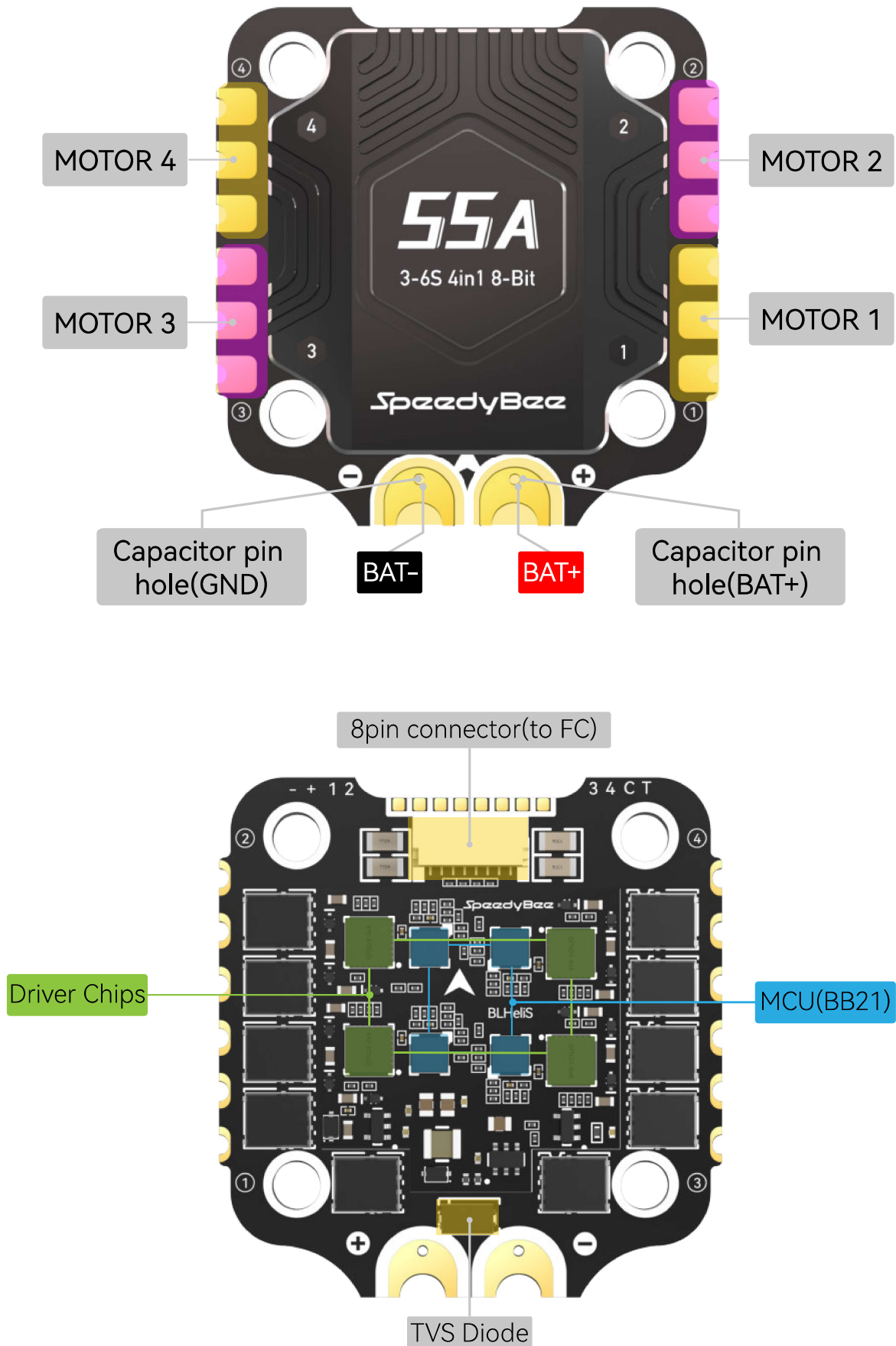


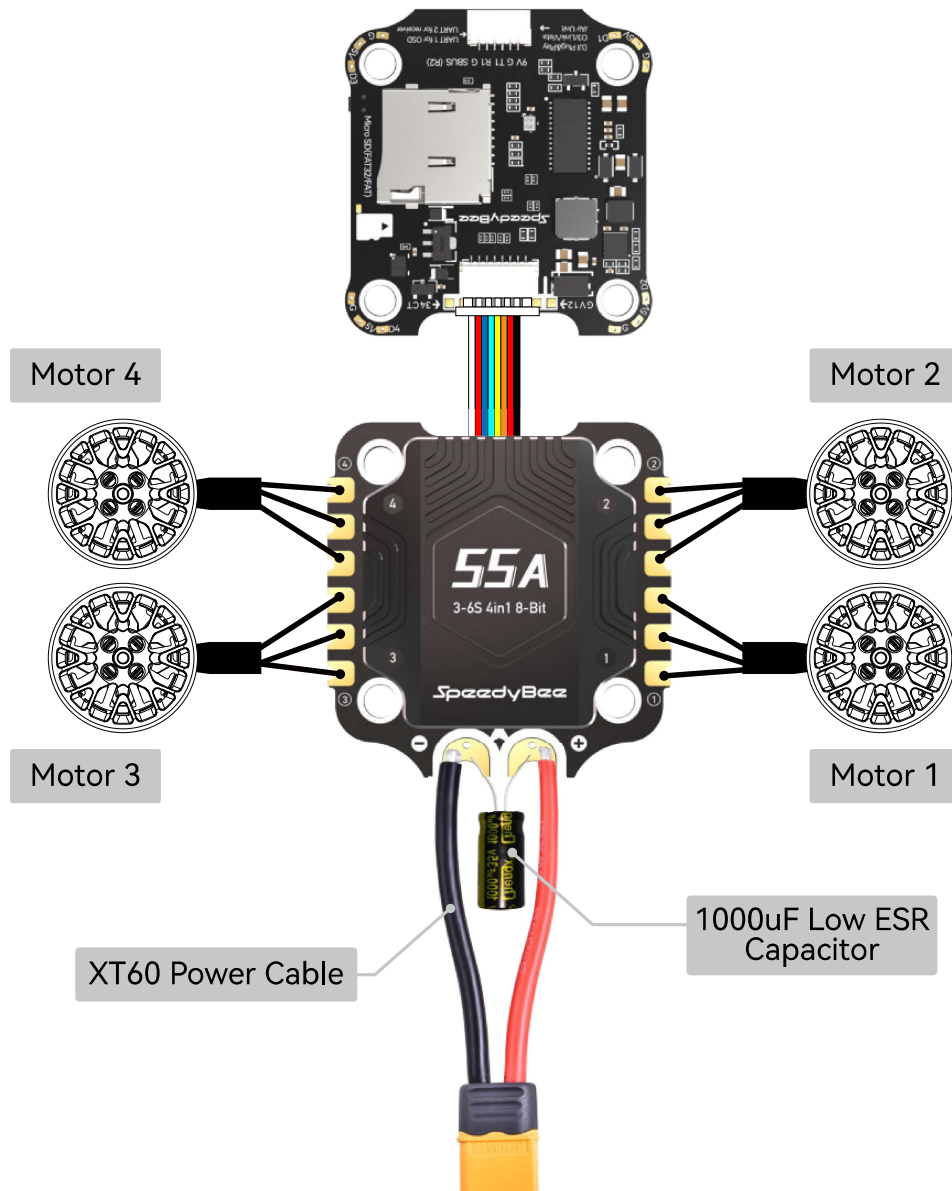
Product Name	SpeedyBee F405 V4 30x30 Flight Controller
MCU	STM32F405
IMU(Gyro)	ICM42688P
USB Port Type	Type-C
Barometer	Built-in
OSD Chip	AT7456E chip
BLE Bluetooth	Supported. Used for Flight Controller configuration (MSP should be enabled with Baud rate 115200 on UART4)
WIFI	Not supported
DJI Air Unit Connection Way	Two ways supported: 6-pin connector or direct soldering.
6-pin DJI Air Unit Plug	Supported. Completely compatible with DJI O3/RunCam Link/Caddx Vista/DJI Air Unit V1, no wire is needed to be changed.
Blackbox MicroSD Card Slot	*Betaflight firmware requires the type of the microSD card to be either Standard (SDSC) or High capacity (SDHC) under 32GB, so extended capacity cards (SDXC) are not supported (Many high-speed U3 cards are SDXC). Also the microSD card MUST be formatted with the FAT16 or FAT32 (recommended) format. So, you could use any SD card less than 32GB, but the Betaflight can only recognize 4GB maximum. We suggest you use this 3rd party formatting tool and choose 'Overwrite format' then format your card. Also check out here for the recommended SD cards or buy the tested cards from our store.
Current Sensor Input	Supported. For SpeedyBee BLS 55A ESC, please set scale = 400 and Offset = 0.
Power Input	3S - 6S Lipo(Through G, BAT pins/pads from the 8-pin connector or 8-pads on the bottom side)
5V Output	9 groups of 5V output, four +5V pads and 1 BZ+ pad(used for Buzzer) on front side, and 4x LED 5V pads. The total current load is 3A.
9V Output	2 groups of 9V output, one +9V pad on front side and other included in a connector on bottom side. The total current load is 3A.
3.3V Output	Supported. Designed for 3.3V-input receivers. Up to 500mA current load.
4.5V Output	Supported. Designed for receiver and GPS module even when the FC is powered through the USB port. Up to 1A current load.
ESC Signal	M1 - M4 on bottom side and M5-M8 on front side.
UART	6 sets(UART1, UART2, UART3, UART4(Dedicated for Bluetooth connection)), UART5 (Dedicated for ESC telemetry),UART6
ESC Telemetry	UART R5(UART5)
I2C	Supported. SDA & SCL pads on front side. Used for magnetometer, sonar, etc.
Traditional Betaflight LED Pad	Supported. 5V, G and LED pads on bottom of the front side. Used for WS2812 LED controlled by Betaflight firmware.
Buzzer	BZ+ and BZ- pad used for 5V Buzzer
BOOT Button	Supported.
	[A]. Press and hold BOOT button and power the FC on at the same time will force the FC to enter DFU mode, this is for firmware flashing when the FC gets bricked.
	[B]. When the FC is powered on and in standby mode, the BOOT button can be used to controller the LED strips connected to LED1-LED4 connectors on the bottom side. By default, short-press the BOOT button to cycle the LED displaying mode. Long-press the BOOT button to switch between SpeedyBee-LED mode and BF-LED mode. Under BF-LED mode, all the LED1-LED4 strips will be controlled by Betaflight firmware.
RSSI Input	Supported. Named as RS on the front side.
Smart Port / FPort	Not supported
Supported Flight Controller Firmware	BetaFlight(Default), INAV
Firmware Target Name	SPEEDYBEEF405V4
Mounting	30.5 x 30.5mm(4mm hole diameter)
Dimension	41.6(L) x 39.4(W) x 7.8(H)mm
Weight	10.5g

Part 3 - SpeedyBee BLS 55A 4-in-1 ESC

Layout

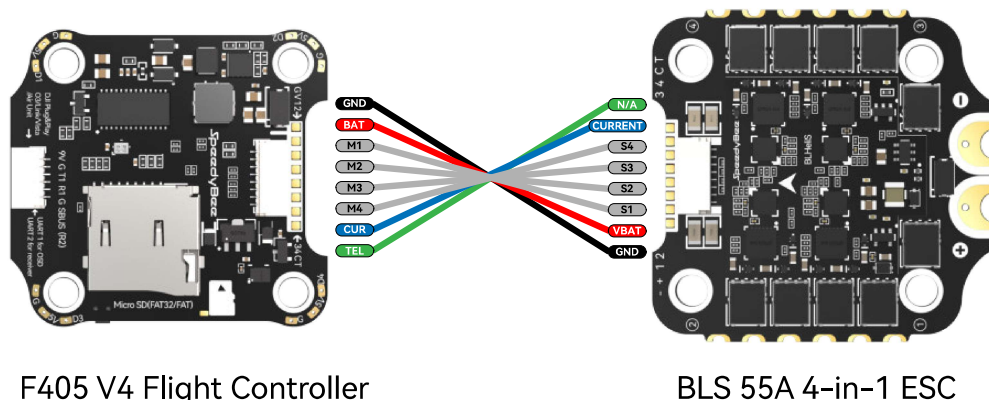
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Note1: In order to prevent the stack from being burnt out by voltage spikes on powering up, it is strongly recommended to use the Low ESR capacitor in the package.

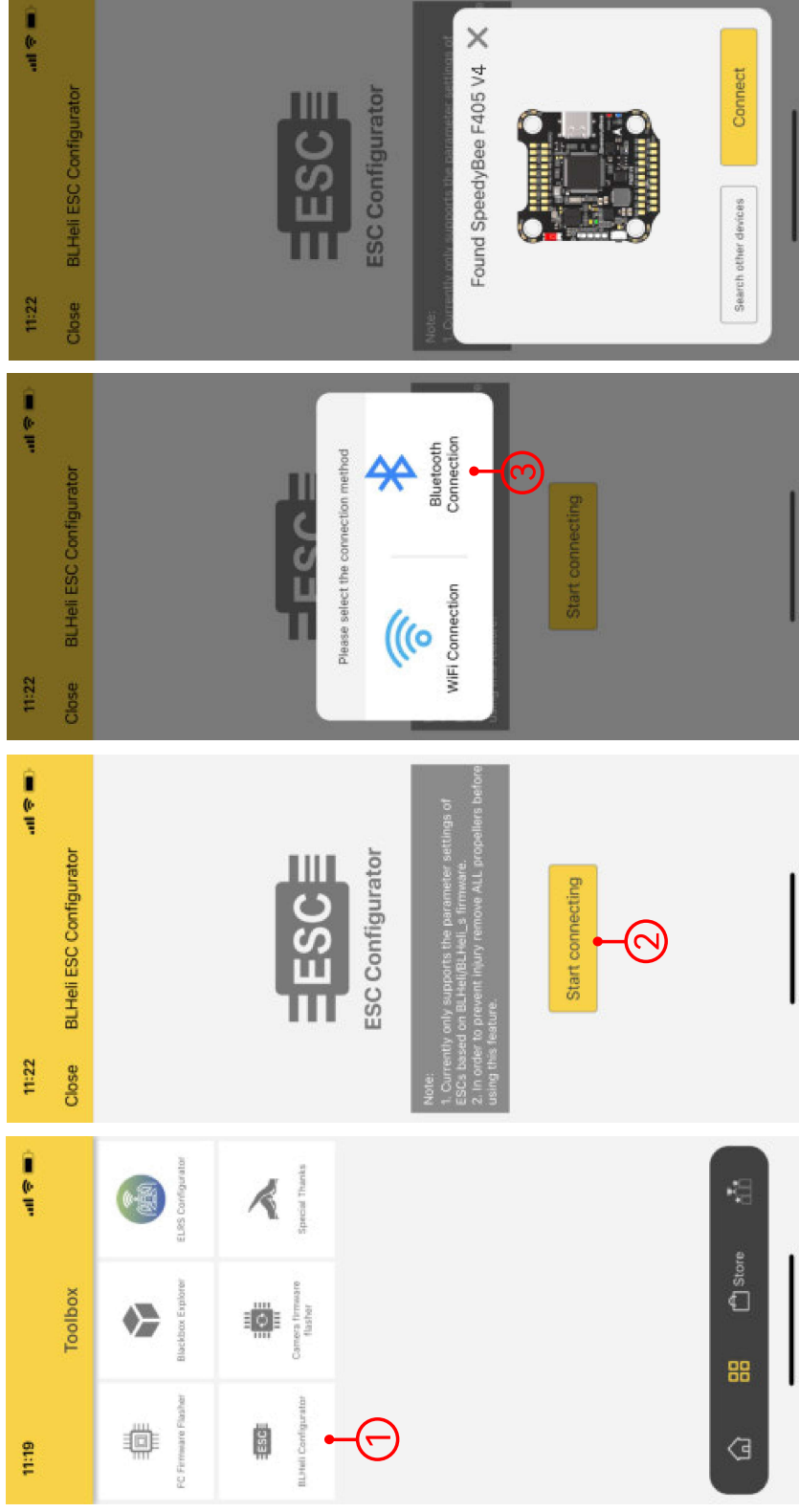
Note2: The FC and ESC can also be connected via direct soldering. Soldering pads definition is as follows.



ESC Configuration

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- You could use the SpeedyBee APP to configure this ESC wirelessly for both BLHeli_S or Bluejay firmware. Steps:



- You could also use PC configurators to configure this ESC. We recommend the ESC Configurator. Please use Google Chrome browser and visit: <http://www.esc-configurator.com>.

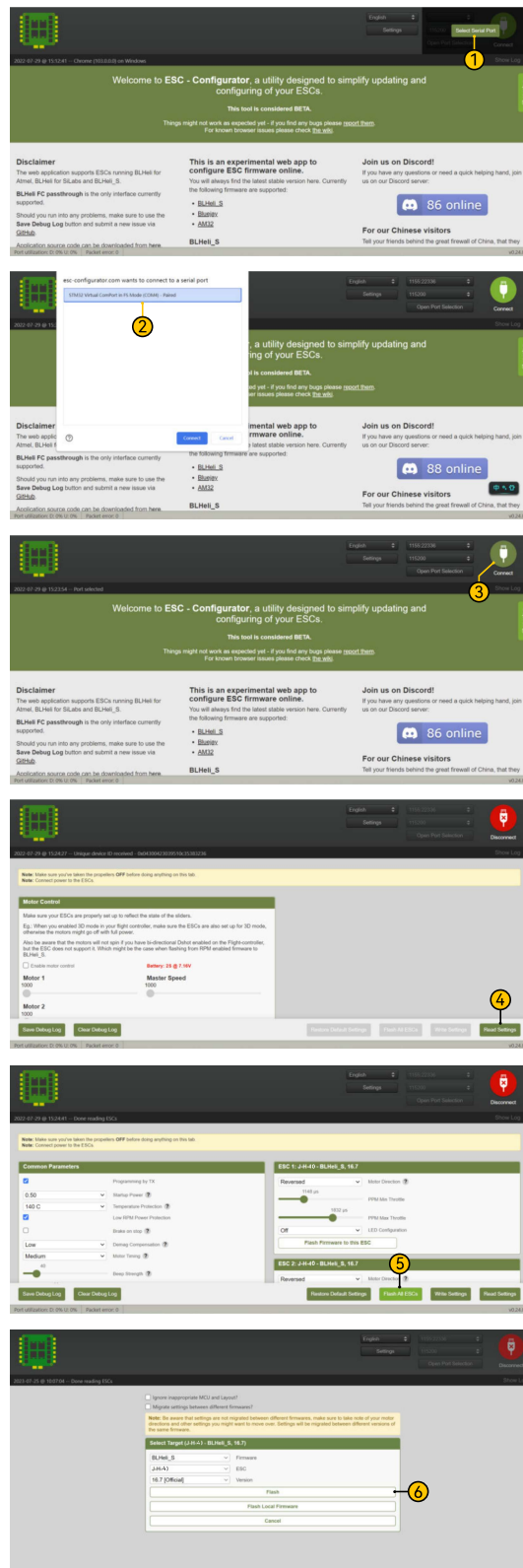
- This 8-bit 55A ESC can run BLHeliS or Bluejay firmware. It is loaded with BLHeliS firmware by default. You could also flash it to Bluejay firmware which can support RPM filtering and Bi-directional Dshot.

Firmware flashing steps are as follows:

- Remove all the propellers from your drone.
- Ensure that the flight controller is connected properly to the ESC, then power up the drone. This step ensures that the ESC starts up correctly.
- Connect the flight controller to the computer using a USB Type-C cable.
- Open the Chrome browser and visit the following website: <https://www.esc-configurator.com/>
- Follow the firmware flashing steps as shown in the screenshots below.

Important note:

On the 6th interface, the "ESC" type must be selected as "J-H-40".



Specifications

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Product Name	SpeedyBee BLS 55A 30x30 4-in-1 ESC
Firmware	BLHelj_S J-H-40
PC Configurator Download Link	https://esc-configurator.com/
Continuous Current	55A * 4
Burst Current	70(10 seconds)
TVS Protective diode	Yes
External Capacitor	1000uF Low ESR Capacitor(In the package)
ESC Protocol	DSHOT300/600
Power Input	3-6S LiPo
Power Output	VBAT
Current Sensor	Support (Scale=400 Offset=0)
ESC Telemetry	Not supported
Mounting	30.5 x 30.5mm(4mm hole diameter)
Dimension	45.6(L) * 44(W) *8mm(H)
Weight	23.5g

pixhawk[®] 4

The most advanced development kit for the PX4® autopilot

Product features

- A new and small form factor
- More computing power and 2X the RAM than previous versions
- New sensors with higher temperature stability
- Integrated vibrations isolation
- Increased ease-of-use: pre-installed with most recent PX4 (v1.8)
- Additional ports for better integration and expansion



Product description

Pixhawk® 4 is the latest update to the successful family of Pixhawk flight controllers. It is designed and developed in collaboration with Holybro and the PX4 team, optimized to run the full Dronecode stack and comes preinstalled with the latest PX4 firmware (v1.8).

It features the currently most advanced processor technology from STMicroelectronics®, sensor technology from Bosch®, InvenSense®, and a NuttX real-time operating system, delivering incredible performance, flexibility, and reliability for controlling any autonomous vehicle.

The Pixhawk 4's microcontroller now has a 2MB flash memory and 512KB RAM. With the increased power and RAM resources, developers can be more productive and efficient with their development work. More complex algorithms and models can be implemented on the autopilot.

High-performance, low-noise IMUs on board are designed for stabilization applications. Data-ready signals from all sensors are routed to separate interrupt and timer capture pins on the autopilot, permitting precise time-stamping of sensor data. Newly designed vibration isolations enables more accurate readings, allowing vehicles to reach better overall flight performances.

The two external SPI buses and six associated chip select lines allow to add additional sensors and SPI-interfaced payload. There are total of four I2C buses, two dedicated for external use and two grouped with serial ports for GPS/compass modules.

The Pixhawk 4 autopilot development kit is perfect for developers at corporate research labs, startups, and for academics (research, professors, students).



Technical specifications

- Main FMU Processor: STM32F765
 - 32 Bit Arm® Cortex®-M7, 216MHz, 2MB memory, 512KB RAM
- IO Processor: STM32F100
 - 32 Bit Arm® Cortex®-M3, 24MHz, 8KB SRAM
- On-board sensors
 - Accel/Gyro: ICM-20689
 - Accel/Gyro: BMI055
 - Mag: IST8310
 - Barometer: MS5611
- GPS: ublox Neo-M8N GPS/GLONASS receiver; integrated magnetometer IST8310

Electrical data

Voltage Ratings:

- Power module output: 4.9~5.5V
- Max input voltage: 6V
- Max current sensing: 120A
- USB Power Input: 4.75~5.25V
- Servo Rail Input: 0~36V

Mechanical data

- Dimensions: 44x84x12mm
- Weight: 15.8g

Interfaces

- 8-16 PWM servo outputs (8 from IO, 8 from FMU)
- 3 dedicated PWM/Capture inputs on FMU
- Dedicated R/C input for CPPM
- Dedicated R/C input for Spektrum / DSM and S.Bus with analog / PWM RSSI input
- Dedicated S.Bus servo output
- 5 general purpose serial ports
 - 2 with full flow control
 - 1 with separate 1.5A current limit
- 3 I2C ports
- 4 SPI buses
 - 1 internal high speed SPI sensor bus with 4 chip selects and 6 DRDYs
 - 1 internal low noise SPI bus dedicated for Barometer with 2 chip selects, no DRDYs
 - 1 internal SPI bus dedicated for FRAM
 - Supports dedicated SPI calibration EEPROM located on sensor module
 - 1 external SPI buses
- Up to 2 CANBuses for dual CAN with serial ESC
 - Each CANBus has individual silent controls or ESC RX-MUX control
- Analog inputs for voltage / current of 2 batteries
- 2 additional analog inputs

Environmental data, quality & reliability

- Operating temp. -40~85C
- Storage temp. -40~85C
- CE
- FCC
- RoHS compliant (lead-free)

For more information visit:

www.dronecode.org
www.pixhawk.org

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